

Single-cell transcriptomics identifies a dampened neutrophil function and accentuated T-cell cytotoxicity in tobacco flavored e-cigarette exposed mouse lungs

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eLife Assessment

This manuscript by Kaur et al. identifies differential gene expression observed in distinct mouse lung cell populations, namely myeloid and lymphoid cells, upon short-term exposure to e-cig aerosols with various flavors. Their findings are potentially **useful** because the single-cell sequencing data provides a reference for future studies of genes and cellular pathways that are most affected by e-cig aerosols and their components. However, the evidence is **incomplete** due to limited statistical analyses and few biological replicates, as well as a lack of experimental validation.

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Abstract

E-cigarettes (e-cigs) are a public health concern for young adults due to their popularity and evidence for increased oxidative stress and immunotoxicity. Yet an extensive study defining the cell-specific immune changes upon exposure to flavored e-cigs remains elusive. To understand the immunological lung landscape upon acute nose-only exposure of C57BL/6J to flavored e-cig aerosols we performed single-cell RNA sequencing (scRNA seq). scRNA profiles of 71,725 cells were generated from control and treatment groups (n=2/sex/group). A distinct phenotype of Ly6G-neutrophils was identified in lungs exposed to tobacco flavored e-cig aerosol which demonstrated dampened IL-1 mediated and pattern recognition signaling as compared to air controls. Differential gene expression analyses identified dysregulation of T-cell mediated pro-inflammation (*Cct7*, *Cct8*) and stress-response signals (*Neurl3*, *Stap1*, *Cirbp* and *Htr2c*) in the lungs of mice exposed to e-cig aerosols, with pronounced effects for tobacco flavor. Flow cytometry analyses and cytokine/chemokine assessments within the lungs corroborated the scRNA seq data, demonstrating a significant increase in T-cell percentages and levels of T-cell associated cytokine/chemokines in the lungs of tobacco-flavored aerosol exposed mice. Increased levels of *Klra4* and *Klra8* expression also suggest an enhanced natural killer (NK) cell activity in this mouse group. Overall, this is a pilot study identifying increase in the percentages of Ly6G-neutrophils that may be responsible for dampened

innate immune responses and heightened T-cell cytotoxicity in lungs of tobacco-flavored e-cig aerosol exposed mice. In addition, we provide preliminary evidence for sex-specific changes in the transcriptional landscape of mouse lungs upon exposure to e-cig aerosol, an area that warrants further study.

Introduction

Electronic cigarettes (e-cigs) or electronic nicotine delivery systems are a relatively novel set of tobacco and flavored products that have gained immense popularity amongst adolescents and young adults in the many countries including United States (US), United Kingdom (UK) and China. Flavors are one of the key features that make these products alluring to the younger diaspora (1). Reports indicate that in 2020 about 22.5% of high school students and 9.2% of middle school students in the US were daily vapers or e-cig users with fruit (66%), mint (57.5%), and menthol (44.5%) being the most preferred flavors (2). However, not much is known about the flavor-specific effects of e-cig vaping on the health and immunity of an individual.

E-cig products and aerosols are known to contain harmful constituents including formaldehyde, benzaldehyde, acrolein, n-nitrosamines, volatile organic compounds (VOCs), ketenes, and metal ions (3–5). Studies have indicated that exposure to e-cigs may enhance inflammatory responses, oxidative stress, and genomic instability in exposed cells or animal systems (6–8). Risk assessment of the non-carcinogenic (systemic) risks of inhaled diacetyl, a potential component of e-liquids, has estimated the hazard quotient to be greater than 1 amongst teens (9). Furthermore, clinical and *in vivo* studies have suggested that exposure to e-cig aerosols could impair innate immune responses in the host thus making them more susceptible to bacterial/viral infections. The bacterial clearance, mucous production, and phagocytic responses in these individuals are shown to be affected upon use of e-cigs (10–14).

However, cell-specific changes within the lung upon vaping is not fully understood making it hard to determine the long-term health impacts of the use of these novel products. In this pilot study, we aim to determine effects of e-cig exposure on mouse lungs at single cell level. To do so, we exposed C57BL/6J mice to 5-day nose-only exposure to air, propylene glycol: vegetable glycerin (PG:VG), fruit-, menthol- and tobacco-flavored e-cig aerosols. We performed single-cell RNA sequencing (scRNA seq) on the lung digests from exposed and control animals and identified neutrophils and macrophages, among others, as the major cell populations in the lung that were affected upon acute exposure. We were able to identify 29 gene targets that were commonly dysregulated amongst all our treatment groups upon aggregating results from the major lung cell types identified in this study. These gene targets are the markers of early immune dysfunction upon e-cig aerosol exposure *in vivo* and could be studied in detail to understand temporal changes in their expression that may govern allergic responses and adverse pulmonary health outcomes upon acute and sub-acute exposures to e-cigarette aerosols.

Results

Fruit-flavored e-cig aerosol results in lung injurious responses upon acute exposure *in vivo*

This study was designed to characterize the effects of exposure to flavored e-cig aerosol at single-cell level to understand the immunological changes in the lung microenvironment upon e-cig use. To do so, we generated a first ever single-cell profile of e-cig aerosol exposed mouse lungs (n= 2/sex/group). The thus obtained results were then validated with the help of our validation cohort of n= 3/sex/group as shown in **Fig 1A**. Since all the commercially available e-liquids used in this

study contained tobacco derived nicotine, we first determined the level of serum cotinine (a metabolite of nicotine) to prove successful exposure to the mice in each treatment group. As expected, we did not see any traces of cotinine in the serum of air and PG:VG exposed mice. Significant levels of cotinine were detected in the serum of mice exposed to fruit-, menthol-, and tobacco-flavored e-cig aerosols (**Fig 1C** [↗](#)).

Since metals released upon heating of the coils of e-cig devices are a source of toxicity upon vaping ([15](#) [↗](#), [16](#) [↗](#)), we further monitored the levels of metals in the e-cig aerosols generated during each day of mouse exposures. This acted as an indirect measure of characterizing the chemical properties of the aerosols used for exposure in this study. To monitor the release of metals into the lungs of the animals, the condensate from each day of exposure was collected and the levels of select elements were detected using ICP-MS. A detailed account of the concentrations of identified elements/metals is provided in **Table 1** [↗](#). Interestingly, we identified flavor-dependent changes in the levels of metals like Ni, Zn, Na, K and Cu. Of note, despite the use of same wattage and temperature (max of 230°C) for generation of e-cig aerosols, the leaching of each metal varied per day of exposure (**Fig 1D** [↗](#)). This is a crucial result as it highlights the importance of studying the impacts of coil composition and device design on the chemical composition of generated e-cig aerosols. These variations might affect the risk and toxicity associated with each of these products, an area that warrants further study. Next, we performed H&E staining of the lung tissue sections to study the morphometric changes in the mouse lungs upon exposure to differently flavored e-cig aerosols. We did not find much evidence of tissue damage or airspace enlargement upon acute exposures in our model as expected. However, we found evidence of increased alveolar septa thickening in the lungs of both male and female mice exposed to fruit-flavored e-cig aerosols (**Figure 1E** [↗](#)). However, this could be a result of lower agarose inflation observed for the mouse lungs in this group of mice. We had challenges/difficulties with inflating the mouse lungs for all the mice in this group, however male mice showed more restriction than the female. Owing to the lack of proper inflation, the lungs were in a collapsed state than the other groups where the lungs were fully inflated. We have also performed mean linear intercept (Lm) measurements for all the lung tissue sections to show lowered Lm values for mouse lungs exposed to fruit-flavored e-cig aerosols as compared to air controls (**Figure 1F** [↗](#)). Airway obstruction could be indicative of “atelectasis”, a condition that may result from lack of surfactant or alveolar damage thus affecting the lung expansion in these mice. Since this was not the prime focus of this study, we did not conduct further experiments to confirm our speculations, but nevertheless, we report these changes to inform and encourage further research in this area.

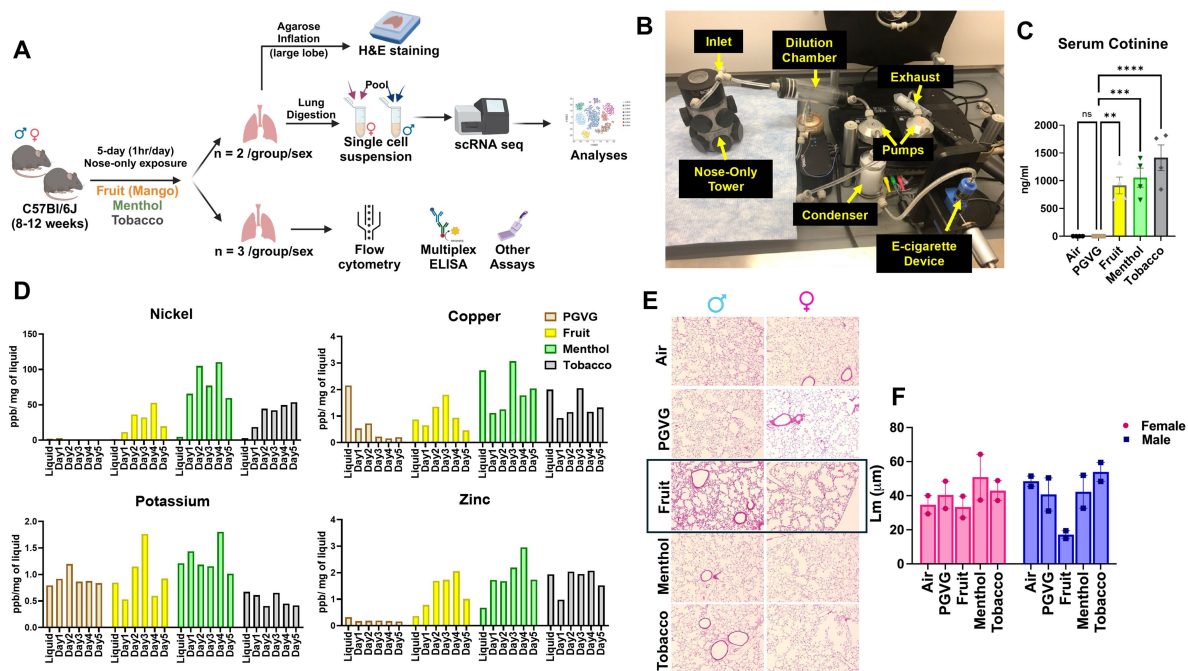


Figure 1

E-cig aerosols show flavor-dependent changes in the levels of quantified metals

Schematics showing the exposure profile and experimental design to understand the effect of exposure to differently flavored e-cig aerosols in the lungs from C57BL/6J mice using scRNA seq (A) and the nose only exposure system used for performing the mouse experiment (B). The exposure characteristics were assessed by measuring the serum cotinine levels in the blood of exposed and control mice. Data are shown as mean $\pm$ SEM ($n = 4/\text{group}$); ns: not significant. ** $p < 0.01$, and *** $p < 0.001$ vs air, per one-way ANOVA for multiple comparison (C) and analyzing the levels of metals in the aerosols captured from each day of exposure using ICP-MS (D). Lung morphometric changes as observed using H&E staining of lung slices from air, PGVG and differently flavored e-cig exposed mice. Representative images of $n = 2/\text{sex}/\text{group}$ at 10X magnification is provided (E) and quantified values for mean linear intercept (Lm) plotted in a sex-specific manner (F)

Element	Eliquid; ppb/ mg of eliquid				E-cig Aerosol (Mean $\pm$ SD); ppb/ mg of eliquid			
	PGVG	Fruit	Menthol	Tobacco	PGVG	Fruit	Menthol	Tobacco
S	75.63	68.36	95.34	79.79	70.67 $\pm$ 16.94	74.05 $\pm$ 39.63	81.31 $\pm$ 27.24	91.81 $\pm$ 11.67
Ni	2.09	1.63	4.73	2.84	1.04 $\pm$ 1.00	30.47 $\pm$ 14.28	83.52 $\pm$ 20.53	41.73 $\pm$ 12.25
Cu	2.16	0.87	2.72	2.00	0.37 $\pm$ 0.22	1.04 $\pm$ 0.48	1.85 $\pm$ 0.70	1.32 $\pm$ 0.39
Si	1.05	1.13	1.55	1.26	1.10 $\pm$ 0.14	1.09 $\pm$ 0.35	1.32 $\pm$ 0.28	1.33 $\pm$ 0.12
K	0.80	0.84	1.21	0.67	0.94 $\pm$ 0.13	0.99 $\pm$ 0.45	1.32 $\pm$ 0.28	0.51 $\pm$ 0.10
Na	0.39	0.40	1.10	0.63	0.65 $\pm$ 0.11	0.91 $\pm$ 0.17	1.70 $\pm$ 0.37	0.79 $\pm$ 0.11
W	0.48	0.23	0.24	0.14	0.46 $\pm$ 0.20	0.30 $\pm$ 0.18	0.28 $\pm$ 0.14	0.22 $\pm$ 0.05
Zn	0.32	0.36	0.68	1.94	0.18 $\pm$ 0.01	1.45 $\pm$ 0.48	2.06 $\pm$ 0.48	1.71 $\pm$ 0.42

Ir	0.28	0.47	0.27	0.11	1.19 ± 0.76	0.35 ± 0.29	0.19 ± 0.12	0.14 ± 0.05
B	0.14	0.02	0.02	0.02	0.07 ± 0.02	0.02 ± 0.01	0.02 ± 0.01	0.02 ± 0.01
Ta	0.14	0.34	0.24	0.13	0.73 ± 0.30	0.24 ± 0.09	0.18 ± 0.06	0.13 ± 0.02
Hf	0.13	0.18	0.13	0.07	0.43 ± 0.19	0.14 ± 0.05	0.09 ± 0.03	0.07 ± 0.01
Mo	0.08	0.01	0.01	0.01	0.02 ± 0.01	0.07 ± 0.01	0.15 ± 0.04	0.06 ± 0.02
Pd	0.07	0.14	0.11	0.06	0.30 ± 0.14	0.11 ± 0.06	0.07 ± 0.02	0.06 ± 0.01
Pt	0.05	0.06	0.05	0.07	0.16 ± 0.07	0.09 ± 0.05	0.08 ± 0.04	0.08 ± 0.03
Zr	0.04	0.01	0.02	0.02	0.02 ± 0.01	0.02±0.0 1	0.01 ± 0.00	0.01 ± 0.00
Sn	0.02	0.06	0.05	0.04	0.06 ± 0.02	0.07 ± 0.02	0.03 ± 0.01	0.03 ± 0.01
Mg	0.01	0.01	0.04	0.03	0.02 ± 0.00	0.12 ± 0.03	0.12 ± 0.03	0.10 ± 0.02
Al	0.01	0.01	0.01	0.01	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00
Co	0.01	0.01	0.01	0.01	0.01 ±	0.02 ±	0.03 ±	0.02 ±
					0.00	0.00	0.01	0.00
Nb	0.01	0.00	0.00	0.00	0.01 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Ba	0.01	0.01	0.01	0.01	0.01 ± 0.01	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00
Re	0.01	0.01	0.01	0.02	0.06 ± 0.04	0.03 ± 0.03	0.02 ± 0.01	0.02 ± 0.01

Table 1

Table showing the levels of common elements found in the flavored e-liquids and e-cig aerosols in as measured using ICP-MS.

Detailed map of cellular composition during acute exposure to e-cig aerosols reveals distinct changes in immune cell phenotypes

As the principle focus of this study was to identify the flavor-dependent and independent toxicities upon exposure to commercially available e-cig aerosols at the single-cell level, we performed the scRNA seq on the mouse lungs from exposed and control mice. After quality control filtering (Figure S1), normalization and scaling, we generated scRNA seq profiles of 71,725 cells in total. Except for the PGVG group, all the rest of the treatments had approximately similar cell viabilities, cell capture, and other quality assessments in this study. However, for uniform analyses, equal features/ genes were used across all the groups for subsequent analyses. A detailed account of the cell viability, number of samples (single-cell capture), and gene features identified before and after filtering upon QC check of sequenced data is provided in Table S1.

Uniform Manifold Approximation and Projection (UMAP) was used for dimensionality reduction and visualization of cell clusters. Cell annotations were performed based on the established cell markers in Tabula Muris database and available published literature. The general clustering of individual cell types based upon the commonly known cell markers was used to identify - **Endothelial** (identified by expression of *Cldn5*), **Epithelial** (identified by expression of *Sfgtpa1*), **Stromal** (identified by expression of *Col3a1*), and **Immune** (identified by expression of *Ptpnc*) cells. The Immune cell population was further subclustered into lymphoid and myeloid populations in the UMAP analyses (Figure 2A & B). The relative cell percentages were plotted in a sex-dependent manner. Of note, upon assessing the sex- and treatment-specific changes in observed cell frequencies for individual cell types using two-way ANOVA, we did not find any significant variation in the relative frequencies of the general cell types in our treatment groups across both sexes (Figure 2C). A detailed account of the two-way ANOVA statistics for individual cell types is provided in Table S2.

DESeq2 was then used to identify the differentially regulated genes in each broad cell category. A list of significant ($p_{adj} < 0.05$) differentially expressed genes (DEGs) was generated using aggregate analysis for both sexes for each treatment group and number of up- and down-regulated genes were plotted in a flavor-dependent manner to get a snapshot of the cell-types and flavor-categories that show most dysregulation upon exposure (Figure 2D). Our data showed dysregulation of genes in endothelial, epithelial, and stromal cell populations, but maximum effect was observed on the cells from immune cell clusters. This is not surprising, as the immune system especially the myeloid cells form the frontline of the host's defenses against external stressors (17, 18). Compared to air, we observed a dysregulation of 553 genes (338 upregulated; 215 downregulated) in the myeloid cell cluster of mouse lungs exposed to tobacco-flavored e-cig aerosol. We identified 324 and 24 DEGs in the myeloid lung cluster from mouse lungs exposed to menthol- and fruit-flavored e-cig aerosols respectively as compared to air control. For the lymphoid cluster, we observed maximum dysregulation in the lungs exposed to fruit-flavored e-cig aerosols with a total of 112 DEGs. In contrast, 41 and 9 significant DEGs were identified for lymphoid cluster from lungs exposed to tobacco and menthol-flavored aerosols respectively.

It is important to mention here that most of the commercially available e-liquids /e-cig products use PG:VG as the base liquid to generate the aerosol and act as a carrier for flavoring chemicals. Thus, we compared the effect of PG:VG alone in our study to make further comparisons between

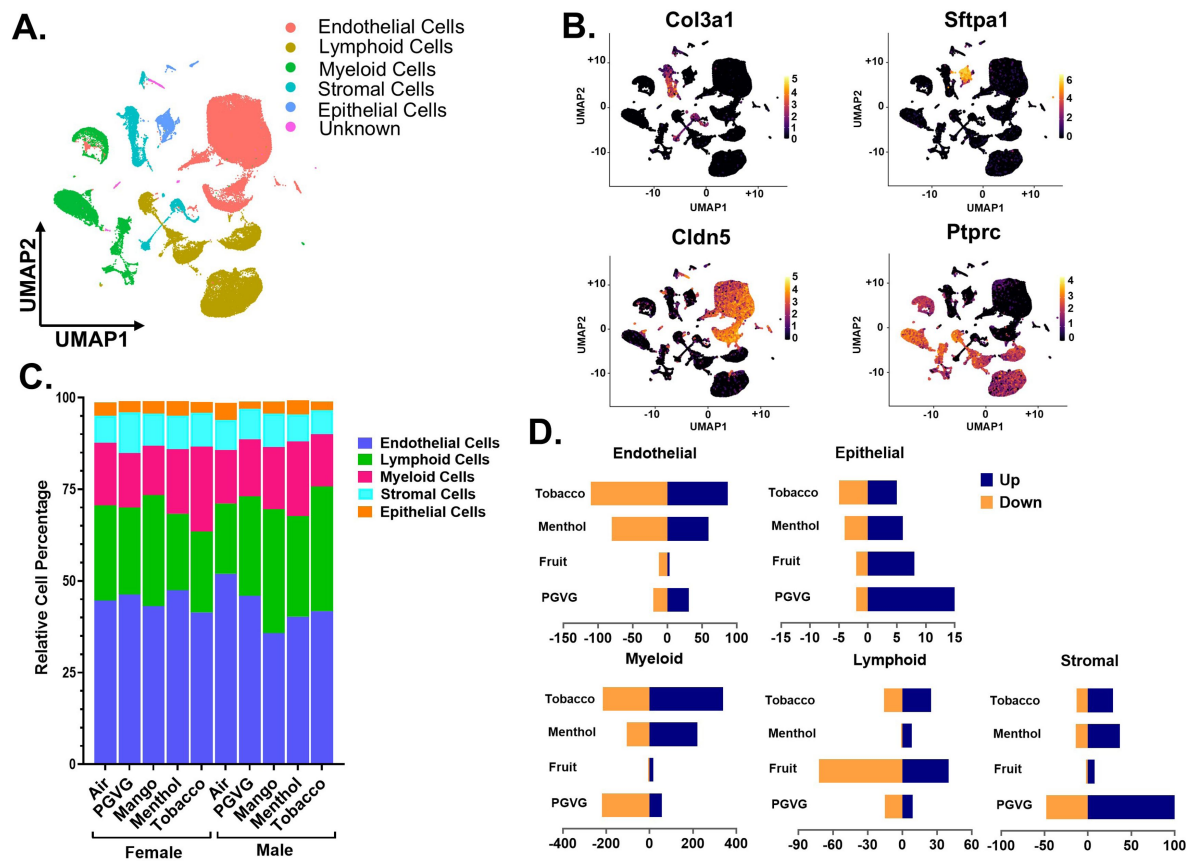


Figure 2

scRNA seq analyses reveals maximum changes in the cell states of immune cell population upon exposure to differently flavored e-cig aerosols.

Male and female C57BL/6J mice were exposed to 5-day nose-only exposure to differently flavored e-cig aerosols. The mice were sacrificed after the final exposure and lungs from air (control) and differently flavored e-cig aerosol (fruit, menthol, and tobacco)-exposed mice were used to perform scRNA seq. UMAP plot of 71,725 cells captured during scRNA seq showing the four major cell populations identified from control and experimental mouse lungs (A) and the expression of canonical markers used for identifying stromal (*Col3a1*), epithelial (*Sftpa1*), endothelial (*Cldn5*) and immune (*Ptprc*) cell populations. The intensity of expression is indicated by the red-yellow coloring (B). The cell frequencies (plotted as percentage) of different cell clusters in each sample type showing the sex-dependent variations in the cell composition on exposure to differently flavored e-cig aerosols (C). Bar graphs showing the number of significant DEGs (p < 0.01) found to be up (blue) and down (orange)-regulated in each cell cluster in differently flavored e-cig aerosol exposed mouse lungs as compared to air controls as determined by aggregate analyses for both sexes in each group using DESeq2 (D).

individual flavoring products with that of PG:VG only. DESeq2 analyses showed dysregulation of 276 genes in mouse lungs exposed to PG:VG alone in the myeloid cell cluster as compared to air controls. Contrary to this, exposure to PG:VG aerosols affected 24 genes in the lymphoid cluster as compared to air control. Furthermore, when compared to PG:VG, exposure to fruit-flavored e-cig aerosol dysregulated 262 genes in the myeloid cluster and 37 genes in the lymphoid cluster (**Table 2**).

Exposure to tobacco-flavored e-cig aerosols dampens

IL-1 mediated signaling in Ly6G-neutrophil cluster

We found maximum dysregulation in immune cell population, more specifically the myeloid cells, upon exposure to e-cig aerosols through our previous analyses. We thus subclustered the myeloid and lymphoid cell populations to further sub-divide their cell types and identify the flavor-dependent changes in immune cell responses. Upon sub clustering the myeloid cell population, we identified 14 unique clusters comprising all the major cell phenotypes including neutrophils, alveolar macrophages (AM), interstitial macrophages (IM), monocytes, dendritic cells (DC), and mast cells (**Figure 3A**). A detailed account of all the cell types identified with their respective marker genes is provided in **Table 3**. Cell cycle scoring revealed the neutrophil cluster to be in G2M phase (**Figure 3B**), indicating that these cell types are either at a checkpoint for uncontrolled proliferation or advancing toward a state of exhaustion (19, 20). In either case, we believed that studying these cells might be of interest to understand the mechanisms governing the immune dysregulation upon exposure of C57BL/6J mice to differently flavored e-cig aerosols. On deeper evaluation, we identified two unique phenotypes of neutrophils (identified by *S100a8*, *Cxcl2*, *Sell* and *Lcn2*) in the mouse lungs. These clusters were named as Ly6G⁺ Neutrophils and Ly6G⁻ Neutrophils based on the presence or absence of Ly6G marker, respectively (**Figure 3C**). Ly6G is important for neutrophil migration and is crucial to regulate neutrophil function within the lung (21). Interestingly we found a consistent flavor-dependent increase (though not significant) in the Ly6G⁻ neutrophil percentage in the lungs of exposed mice as compared to air controls. To validate the changes in cell percentages upon flavor exposure from scRNA seq, we performed flow cytometry. Our data from flow cytometry corroborated with the scRNA analyses results showing an increase in the levels of Ly6G⁺ neutrophils in lung digests from mouse lungs exposed to tobacco-flavored e-cig aerosols. The observed increase was more pronounced in males as compared to females thus highlighting the sex-dependent nature of changes in immune cell frequencies within the exposed lungs (**Figure 3E**).

AM form the bulk of the myeloid cell population in the lung and are characterized by top markers like *Chil3*, *Plet1*, *Lpl* and *Marco* (**Figure 3C**) (22). A flavor-dependent decrease in the resident AM (RAM) population was observed for both menthol and tobacco flavored e-cig aerosol exposed lungs (**Figure 3D**). These changes were again corroborated through the significant decrease in the BALF macrophage cell percentages in menthol-flavored e-cig aerosol exposed mice lungs as compared to air controls for both male and female mice (**Figure 3F**). Of note, though not significant, we saw a decrease in the levels of resident macrophages in the lung digests from mouse lungs exposed to menthol flavored e-cig aerosols with a more pronounced decrease observed amongst female mice (**Figure S2A**). Interestingly we found a distinct population of RAM expressing markers including *Mki67*, *Pclaf* and *Top2a*. These genes are responsible for cell proliferation. Importantly, the cell scoring for the G2M phase of the cell cycle was high for this cell cluster and thus these were labeled as 'proliferating' RAMs (pRAM) (23, 24). The pRAMs showed significant ($p=0.0156$) changes in the cell percentages with respect to varying exposures in our study (**Figures 3D**, **Table S2**).

Taken together, we show a significant shift towards the neutrophil-mediated immune response in the lungs of animals exposed to tobacco-flavored e-cig aerosols in our study. It is important to mention that due to the presence of nicotine in all the e-liquids used as treatment groups for this

Comparisons	Myeloid Cells		Lymphoid Cells		Endothelial cells		Stromal Cells		Epithelial Cells	
	Up	Down	Up	Down	Up	Down	Up	Down	Up	Down
PGVG vs Air	57	219	9	15	31	20	106	48	17	2
Menthol vs Air	220	104	8	1	59	80	37	14	6	4
Fruit Vs Air	17	7	40	72	3	12	8	2	8	2
Tobacco vs Air	338	215	25	16	87	110	29	13	5	5
Menthol vs PGVG	673	200	55	9	38	87	8	22	4	40
Fruit vs PGVG	135	127	25	12	25	20	29	9	12	12
Tobacco vs PGVG	852	408	90	12	52	87	44	48	5	15

Table 2

DESeq 2 results showing the number of significantly ($p_{adj} < 0.05$) up- or downregulated genes in different cell clusters

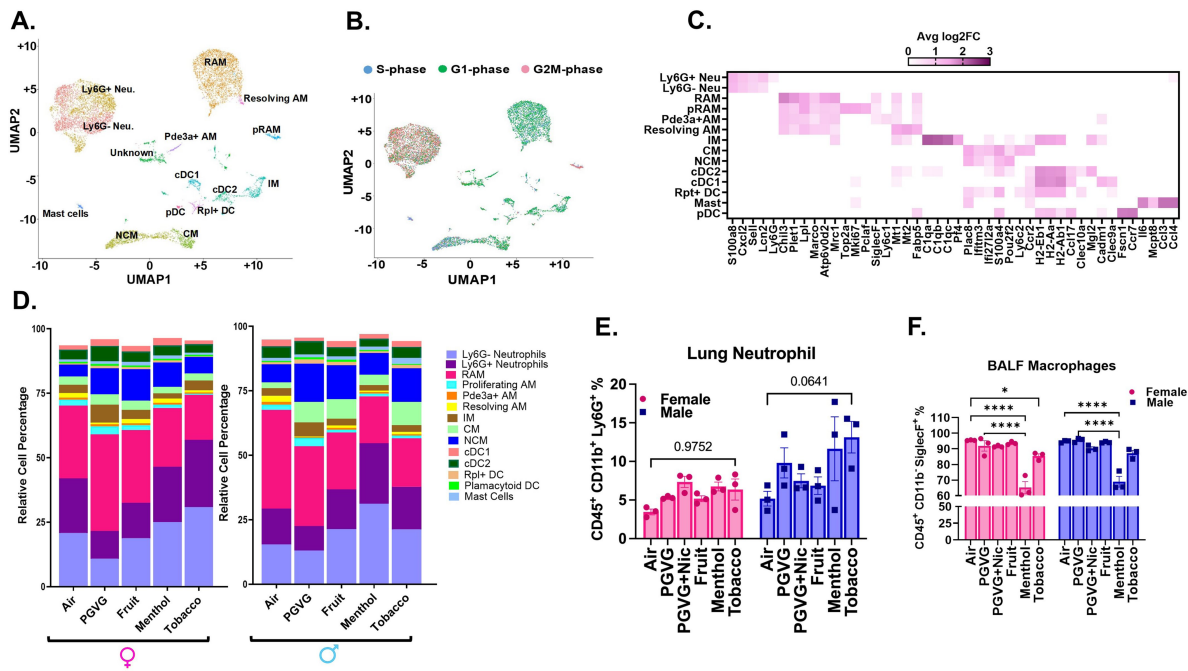


Figure 3

Cellular composition of myeloid cells in air and e-cig aerosol exposed mouse lung reveal a unique population of neutrophils.

A total of 14 unique cell clusters were identified from the myeloid cell subset of air and e-cig aerosol exposed mouse lung (A). UMAP plot showing the cell cycle state of different cell types in the myeloid cluster as determined using the Cell cycle scoring in Seurat (B). The heatmap showing the most DEGs in each cluster across the myeloid subset. These differential expressions were used to determine and name the cell types in this cluster. The intensity of expression is indicated by the purple coloring (C). The cell percentages of different cell types in the myeloid cluster showing the sex-dependent variations in the cell composition on exposure to differently flavored e-cig aerosols (D). Flow cytometry results showing sex-dependent changes in the percentages of lung neutrophils (CD45⁺ CD11b⁺ Ly6G⁺) in lung tissue digest (E) and alveolar macrophages (CD45⁺ CD11b⁺ SiglecF⁺) in BALF (F) from mice exposed to differently flavored e-cig aerosols. Data are shown as mean $\pm$ SEM (n = 3/sex/group). *p<0.05, and ****p<0.0001, per Tukey post hoc two-way ANOVA comparison.

Cell Types	Abbreviations	Cell Types	Abbreviations
Resident alveolar macrophages	RAM	Rpl+ dendritic cell	Rpl+ DC
Proliferating Alveolar macrophages	pRAM	Plasmacytoid dendritic cell	pDC
Resolving alveolar macrophages	Resolving AM	B-cell progenitors	Pro B cell
Pde3a+ alveolar macrophages	Pde3a+ AM	B-cell precursor	Pre B cell
Interstitial macrophages	IM	Mature B cell	Mature B cell
Classical monocytes	CM	Interferon stimulated B-cell	IFN stim. B cell
Non-classical monocytes	NCM	CD8+ T cell	CD8+ T cell
Dendritic cell 1	cDC1	$\alpha\beta$ T cell	$\alpha\beta$ T cell
Dendritic cell 2	cDC2	Regulatory T cells	Treg cell
$\gamma\delta$ T cell	$\gamma\delta$ T cell	Natural Killer cells	NK cell

Table 3

Table showing the cell types identified from myeloid and lymphoid clusters and their abbreviations

study, we added an extra control group-PGVG+Nic-for select experiments. Flow cytometric analyses revealed slight variations in the lung neutrophils and macrophage percentages observed in PGVG+Nic group as compared to PGVG only. However, none of these changes were significant. Furthermore, the patterns of change observed for the lungs exposed to aerosols from flavored e-liquids were quite distinct from those observed for PGVG+Nic thus proving that the observed changes are not solely due to the presence of nicotine in these treatment groups (**Figures 3E-F**, **S1A**).

Differential gene expression analyses on the myeloid cell cluster identified maximum gene dysregulation in myeloid cell cluster from mouse lungs exposed to tobacco-flavored e-cig aerosols. Interestingly, we did not observe major dysregulation in the gene expression for myeloid cell cluster in fruit-flavored e-cig exposed mouse lungs as compared to air controls. A total of 24 genes (17 upregulated; 7 downregulated) were differentially expressed in the myeloid clusters of lungs of animals exposed to fruit-flavored e-cig aerosols (**Figure 4A**). GO analyses identified terms like ‘myeloid cell differentiation’, ‘gas transport’ and ‘H₂O₂ catabolic process’ as the top hits for this cluster (**Figures 4B**). Importantly, few gene clusters (*Hbb-bs*, *Hbb-bt*, *Hba-a2* and *Hbb-a1*) showed sex-specific changes in the gene expressions in exposed lungs as compared to control. These genes clusters enrich for ‘erythrocyte development’ upon GO analyses and warrant further study.

We demonstrated an upregulation of 220 genes and a downregulation of 104 genes in the lungs of mice exposed to menthol-flavored e-cig aerosol as compared to ambient air. We observed increased expression of inflammatory genes including *Cxcl3*, *Il1r2*, *Stat4*, pointing towards the activation of chemokine-mediated signaling due to exposure to menthol-flavored e-cig aerosol. We also found a significant decrease in the expression of *Edn1* (endothelin1) in the myeloid cells of flavored e-cig aerosols as compared to air controls (**Figure 4C**). These genes play a role in positive regulation of neutrophil recruitment and inflammatory responses into the lungs (25–29). GO analyses of the DEGs (p<0.01), thus, showed ‘cytokine-mediated signaling’, ‘cell chemotaxes’, and ‘negative regulation of MAPK cascade’ as the top hits as shown in the **Figure 4D**.

Like menthol-, tobacco-flavored e-cig aerosol also elicited a significant increase in the expression of 338 genes and a decrease in 215 genes as compared to air controls in the myeloid cell cluster. We observed an increase in the expression of chemokines like *Stat4*, *Il1b*, and *Il1bos* in the myeloid cells resulting in terms like ‘cytokine-mediated signaling pathway’ enriched upon gene enrichment analyses. Like menthol-exposed mouse lungs, we found significant downregulation of *Edn1* gene in the lungs of mice exposed to tobacco-flavored e-cig aerosol (**Figure 4E**). We provide the list of all DEGs as obtained from DESeq2 analyses and GO analyses results in Supplementary Excel Files 1-2.

Based on previous literature (30, 31) and enrichment of genes responsible for neutrophil chemotaxes in the current work, we reckoned this cell type to be of importance to be looked at separately *in vivo*. As mentioned earlier, we identified two distinct phenotypes of neutrophils-(a) Ly6G⁺ neutrophils and (b) Ly6G[−] neutrophils. Interestingly, the percentage increase in the Ly6G[−] neutrophil population of aerosol exposure was higher as compared to that for Ly6G⁺ neutrophils per scRNA cell frequency analyses.

Upon performing DESeq2, we did not find any changes in the gene expressions for any of the two neutrophil populations for mouse lungs exposed to menthol-flavored e-cig aerosols despite an increase in the cell percentages of Ly6G[−] neutrophil phenotype. Interestingly, exposure to fruit-flavored e-cig aerosol caused most changes in the gene expressions in neutrophil population, with Ly6G[−] neutrophils being affected more than the Ly6G⁺ population. We observed a dysregulation of 33 and 25 genes in Ly6G[−] neutrophil cluster from fruit- and PG:VG-aerosol exposed mouse lungs respectively. The Egl-9 Family Hypoxia Inducible Factor 3 (*Egln3*) gene was found to be downregulated upon exposure to e-cig aerosols in a flavor-independent manner. *Egln3* has been

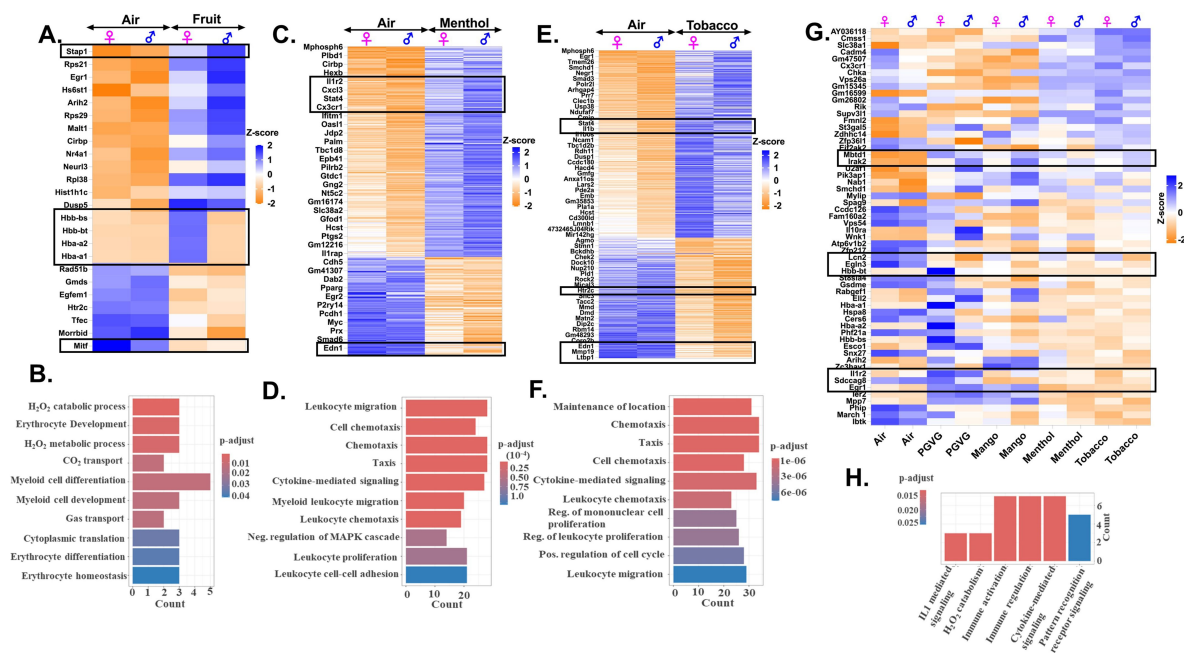


Figure 4

Exposure to flavored e-cig aerosols result in dampening of neutrophil-mediated innate immune responses in C57BL/6J mouse lungs.

Heatmap and bar plot showing the DESeq2 and GO analyses results respectively from the DEGs in the myeloid cell cluster from Fruit (A & B), menthol (C&D) and tobacco (E&F)-flavored e-cig aerosol exposed mouse lungs as compared to air controls. Heatmap (G) and bar plot (H) showing the DESeq2 and GO analyses results respectively from the DEGs in the Ly6G-neutrophil cluster from air and flavored e-cig (fruit, menthol and tobacco) aerosol exposed mouse lungs.

known to play a crucial role in lung cancer and its inactivation exhibits lower tumor burden by induction of cancer senescence (32). We further identified dysregulation in the expressions of genes including *Irak2*, *Egr1*, *Il1r2*, *Pik3ap1*, and *Hspa8* in Ly6G⁺ neutrophils for both fruit- and tobacco-aerosol exposed mouse lungs as compared to air controls. Furthermore, upregulation of genes like *Irak2* and *Egr1* in the Ly6G⁺ neutrophils from mouse lungs exposed to PG:VG and fruit-flavored e-cig aerosol points towards the involvement of NF-κB mediated signaling in our study (Figure 4G). On performing gene enrichment analysis on the dysregulated gene sets from Ly6G⁺ neutrophil cluster we obtained hits including ‘interleukin-1 mediated signaling pathway’, ‘hydrogen peroxide catabolic process’, and ‘pattern recognition signaling pathway’ (Supplementary Excel File 3). Of note, most of the associated genes were upregulated in lungs exposed to fruit-flavored e-cig aerosols. However, the expression pattern of these gene sets points towards dampening of neutrophil-mediated immune responses for lungs exposed to tobacco-flavored e-cig aerosols as compared to air controls (Figure 4H). Compared to Ly6G⁺ neutrophils, we did not observe any change in the gene expression profile of Ly6G⁺ neutrophils for mouse lungs exposed to tobacco-flavored e-cig aerosol. For both, PG:VG and fruit-flavor e-cig aerosol exposed mouse lungs, *Trem14* (Triggering Receptor expressed on Myeloid Cells-Like Protein 4) gene was found to be downregulated in the Ly6G⁺ neutrophils (Figure S2B). This gene functions as a positive regulator of TLR7 signaling (33, 34).

Activation of T-cell cytotoxic responses in lymphoid cells upon exposure to tobacco-flavored e-cig aerosols

We identified 10 clusters upon sub clustering the lymphoid cell population. These comprised of: Pro-B cell, Pre-B cell, Mature B-cell, CD4⁺ T-cell, CD8⁺ T-cell, Treg cell, AB T-cell, GD T-cell and Natural killer (NK) cells (35–37). Based on the expression of *Ifit3*, *Oasl2* and *Rsad2* a unique cluster of IFN-stimulated B-cells was identified. The cell frequency of these cells was markedly lowered (19%) in mouse lungs exposed to tobacco-flavored e-cig aerosols as compared to air controls. Two-way ANOVA analysis of scRNA seq data showed significant changes ($p = 0.0237$) in the cell frequencies of CD4 T cells in a flavor-dependent manner in our treatment groups (Table S2). We also identified sex-dependent changes in the cell frequencies of B-cells ($p = 0.0148$), pro-B cells ($p = 0.0078$) and CD8 T cells ($p = 0.0265$) of the lymphoid cluster in our study (Figures 5A–C, Table S2). Flow cytometric analyses showed a significant increase in the CD4⁺ T-cell percentages in the lung digest from female mice exposed to tobacco-flavored e-cig aerosol. We also observed a significant increase in the CD8⁺ T cell percentages in the lungs of tobacco-flavored e-cig aerosol exposed mice lungs as compared to air control in both male and female mice (Figure 5D). Of note, however, we found significant sex-dependent associations between the CD4⁺ T cell percentages ($p = 0.0339$) upon two-way ANOVA analyses of the flow cytometry data.

We found dysregulation of 112 genes (40 up- and 72-down-regulated) in mouse lungs exposed to fruit-flavored e-cig aerosols in the lymphoid cluster. We observe dysregulation of genes (*Incenp*, *Wnt4*, *Ccnb2*, *Trat1*, *Themis*) enriched for ‘T-cell receptor signaling’ and ‘nuclear division’, in exposed group as compared to air control (Figure 6A–B). Upregulation of genes like *Malt1* (Mucosa-associated lymphoid tissue lymphoma translocation protein 1), *Ppp1r2* (Protein phosphatase inhibitor 2), and *Spib* (Transcription factor Spi-B) indicates activation of T-cell mediated immune response upon exposure to fruit-flavored aerosol (38–40).

Contrary to the responses observed for exposure to fruit-flavored e-cig aerosols, we found significant upregulation in the expression of *Cdk8* and *Camk1d* genes in the lymphoid cell populations for menthol-flavored aerosol exposed mouse lungs (Figure 6C–D). *Cdk8* (cyclin-dependent kinase 8) is a transcriptional regulator that has a role in the cell cycle progression (41). Whereas *Camk1d* (calcium/calmodulin dependent protein kinase ID) functions to regulate calcium-mediated granulocyte function and respiratory burst within the cells (42). Taken together, our results point towards an increase in cell proliferation and gene transcription in the lymphoid cluster of mouse lungs exposed to menthol-flavored e-cig aerosols as indicated by

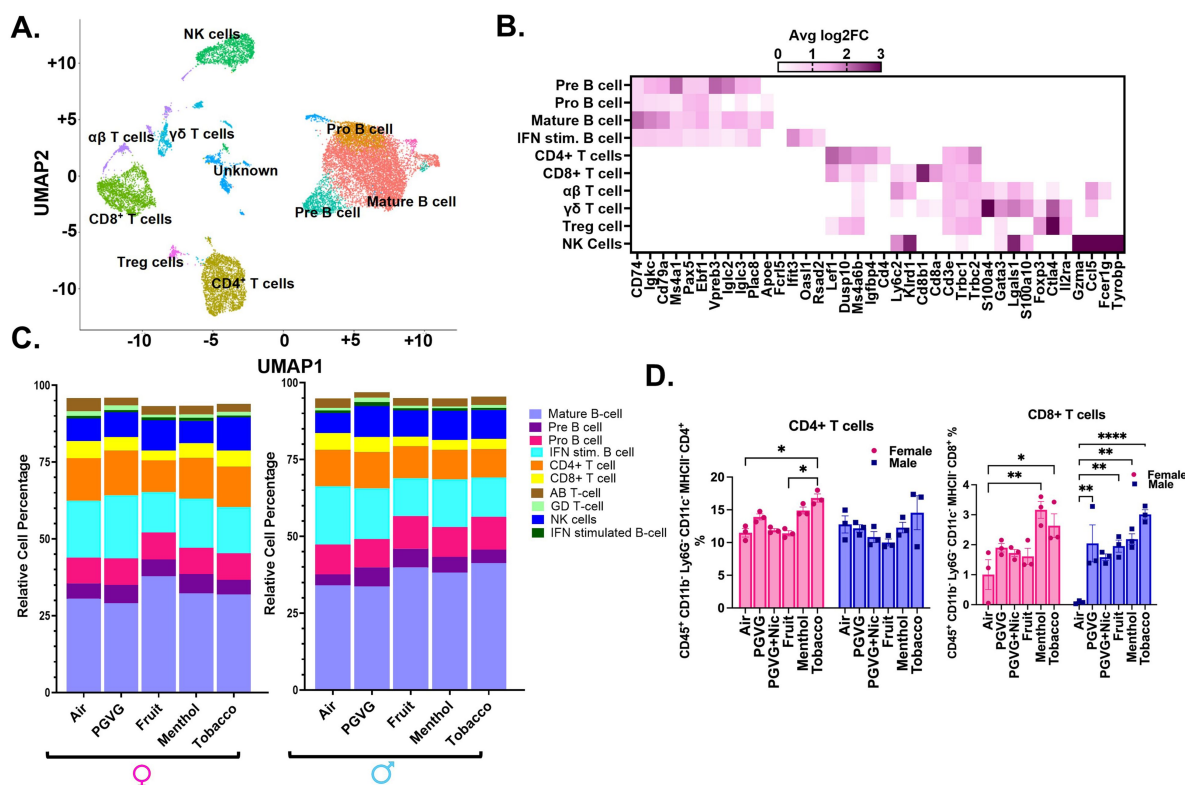


Figure 5

Exposure to tobacco-flavored e-cig aerosols results in significant increase in lung T-cell percentages in C57Bl/6J mice.

A total of 10 unique cell clusters were identified from the lymphoid cell subset of air and e-cig aerosol exposed mouse lung (A). The heatmap showing the most DEGs in each cluster across the myeloid subset. These differential expressions were used to determine and name the cell types in this cluster. The intensity of expression is indicated by the purple coloring (B). The cell percentages of different cell types in the lymphoid cluster showing the sex-dependent variations in the cell composition on exposure to differently flavored e-cig aerosols (C). Flow cytometry results showing changes in the percentages of lung CD4⁺ and CD8⁺ T-cells in the lung tissue digest (D) from mice exposed to differently flavored e-cig aerosols as depicted in a sex-specific manner. Data are shown as mean $\pm$ SEM ($n = 3/\text{sex}/\text{group}$). * $p < 0.05$, ** $p < 0.01$, and **** $p < 0.0001$, per Tukey post hoc two-way ANOVA comparison.

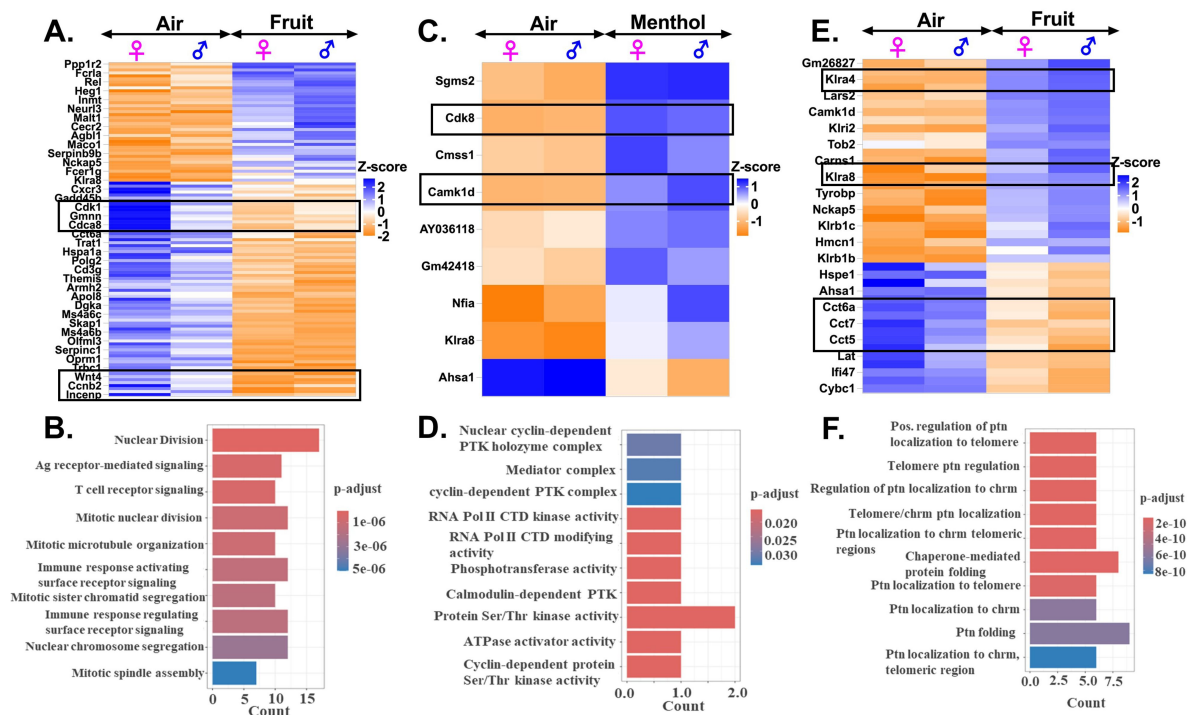


Figure 6

Exposure to tobacco flavored e-cig aerosols result in activation of T-cytotoxic responses in C57BL/6J mouse lungs.

Heatmap and bar plot showing the DESeq2 and GO analyses results respectively from the DEGs in the lymphoid cell cluster from Fruit (A & B), menthol (C&D) and tobacco (E&F)-flavored e-cig aerosol exposed mouse lungs as compared to air controls.

enrichment of terms like ‘cyclin-dependent protein serine/threonine kinase activity’, ‘RNA polymerase II CTD modifying activity’ and ‘calmodulin-dependent protein kinase activity’ associated with this cell cluster.

We observed the downregulation of genes responsible for chaperone-mediated protein folding (*Cct5*, *Cct7*, *Cct8*) in the lymphoid cells from tobacco-flavored e-cig aerosol exposed mouse lungs. Downregulation of these genes could be indicative of the accumulation of misfolded proteins in these lungs which may lead to enhanced cell death (43, 44). In fact, we found increased expression of killer cell lectin-like receptor (*Klra*)-4 and 8 in exposed mice lungs as compared to air controls, thus indicating that upon exposure to tobacco-flavored e-cig aerosols, the lymphoid cells undergo protein misfolding thereby resulting in increased cell death (Figure 6E-F). A detailed account of the DESeq2 results and GO analyses for all the groups is provided in Supplementary Excel files 4-5.

Dysregulation of chemokine signaling and T-cell activation on exposure to flavored e-cig aerosols

Since we showed increased production of cytokines/chemokines, driving the immune responses in mouse lungs exposed to flavored e-cigs, we performed multianalyte assay to determine the levels of these inflammatory cytokines in the lung digests from the exposed animals as shown in Figure 7A. Exposure to tobacco-flavored e-cig aerosol resulted in a marked increase in the levels of chemotactic chemokines including CXCL16, CXCL12, CXCR3 and proinflammatory cytokines including CCL12, CCL17, CCL24, and Eotaxin in the mouse lung digests as compared to air control. Supporting our transcriptional data, the levels of IL1b were dampened in the exposed mouse lungs thus showing negative regulation of IL-1 mediated signaling. We identified sex-dependent changes in the cytokine levels in lung digests from fruit-flavored e-cig aerosols with male mice showing more dampening of cytokine/chemokine protein levels as compared to their female counterparts. Interestingly, the fold changes in the PGVG+Nic group were contrasting to those observed by PGVG alone and flavored e-cig aerosol exposed mouse lungs, again supporting our previous deduction that the observed changes are observed as a concerted effect of chemical present in the e-liquid used for flavored e-cig aerosol exposure.

To identify genes that were commonly dysregulated upon exposure, we generated a list of common genes that were significantly dysregulated in exposure categories (fruit, menthol and tobacco). We identified 9 such target genes-*Neur13*, *Egfm1*, *Stap1*, *Tfec*, *Mitf*, *Cirbp*, *Hist1h1c*, *Gmds*, and *Htr2c* that were dysregulated in the myeloid cluster from lungs exposed to differently flavored e-cig aerosol, but not PGVG. We observed significant upregulation of *Neur13*, *Stap1*, *Cirbp*, and *Hist1h1c* and downregulation of *Tfec*, *Mitf*, *Gmds*, and *Htr2c* in the myeloid cluster of mice exposed to differently flavored e-cig aerosol as compared to air control (Figure 7B). On analyzing the lymphoid cluster for commonly dysregulated genes, we identified - *Klra8* (Killer cell lectin-like receptor 8) and *Nfia* (nuclear factor I)-that were significantly upregulated in the exposure groups as compared to air-controls (Figure 7E). *Klra8* is a natural killer cell associated gene, and its upregulation is generally associated with viral infection associated host immune responses within the mouse lungs (45–47). *Nfia* is a transcriptional activator responsible for regulating Oxphos-mediated mitochondrial responses and proinflammatory pathways (48, 49).

Overall, we identified a total of 29 commonly dysregulated gene targets were identified from five major cell clusters and performed gene enrichment analyses on the identified targets to identify the top hits (Tables 4–5). Terms like ‘negative regulation of immune system’ (*Hmgb3/Gpam/Scgb1a1/Stap1/Ldlr*), ‘positive regulation of lipid biosynthetic pathway’ (*Htr2c/Gpam/Ldlr*), and ‘receptor recycling’ (*Ldlr/Ramp3*) were amongst the top hits in our observations (Figure 7C).

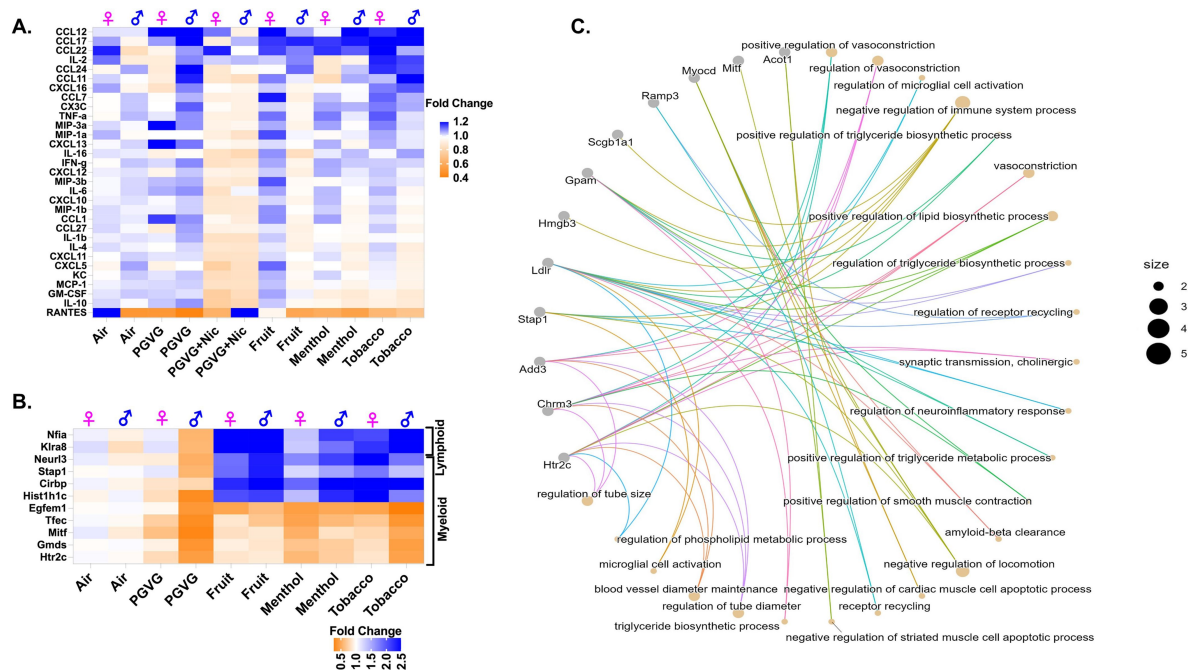


Figure 7

GO analyses of the commonly dysregulated genes show enrichment of dampened innate immune response upon exposure to flavored e-cig aerosols in C57Bl/6J mice.

Heat map showing the sex-dependent average fold changes in the levels of cytokines/chemokines in the lung of differently flavored e-cig aerosol exposed C57Bl/6J mouse lungs as compared to air controls. Data are shown as mean $\pm$ SEM (n = 3/sex/group) (A). Heatmap showing the fold changes in the expression of commonly dysregulated genes in the myeloid cluster and lymphoid clusters in mouse lungs exposed to flavored e-cig aerosols as compared to ambient air as determined after DESeq2 analyses (B). CNET plot results showing the pathways regulated by the common DEGs on acute (5-day) exposure to differently flavored fruit, menthol, and tobacco e-cig aerosols in C57Bl/6J mouse lungs (B).

Genes	Gene Names	Gene Function
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Neurl3	Neuralized E3 Ubiquitin Protein Ligase 3	ubiquitin protein ligase activity
Egfem1	EGF-like and EMI domain containing 1	calcium ion binding activity
Stap1	Signal Transducing Adaptor Family Member 1	protein kinase binding and SH3/SH2 adaptor activity
Tfec	Transcription Factor EC	multiple cellular processes including survival, growth and differentiation
Mitf	Melanocyte Inducing Transcription Factor	critical role in cell differentiation
Cirbp	Cold Inducible RNA Binding Protein	role in cold-induced suppression of cell proliferation
Hist1h1c	H1.2 linker histone	functions in the compaction of chromatin
Gmds	GDP-Mannose 4,6-Dehydratase	coenzyme binding and NADP ⁺ binding
Htr2c	5-Hydroxytryptamine Receptor 2C	G protein-coupled receptor activity
Nfia	Nuclear Factor I A	DNA-binding transcription factor activity
Klra8	killer cell lectin-like receptor	carbohydrate binding activity. Acts upstream of or within response to virus
Trp53i11	Tumor Protein P53 Inducible Protein 11	negative regulation of cell population proliferation
Ehd2	EH Domain Containing 2	Angiopoietin-like protein 8 regulatory pathway and response to elevated platelet cytosolic Ca ²⁺ .
Ackr2	Atypical Chemokine Receptor 2	recruitment of effector immune cells to the inflammation site.
Marcks	Myristoylated Alanine Rich Protein Kinase C Substrate	involved in cell motility, phagocytosis, membrane trafficking and mitogenesis
Pfkl	Phosphofructokinase,	protein binding and monosaccharide binding

Ramp3	Receptor Activity Modifying Protein 3	signaling receptor activity and coreceptor activity
Chrm3	Cholinergic Receptor Muscarinic 3	cellular responses such as adenylate cyclase inhibition, phosphoinositide degeneration, and potassium channel mediation
Sftpa1	Surfactant Protein A1	carbohydrate binding and lipid transporter activity
Add3	Adducin 3	actin binding and calmodulin binding
Hmgb3	High Mobility Group Box 3	important role in maintaining stem cell populations and may be aberrantly expressed in tumor cells.
Acot1	Acyl-CoA Thioesterase 1	involved in acyl-CoA metabolic process; long-chain fatty acid metabolic process; and very long-chain fatty acid metabolic process
H1f0	H1.0 Linker Histone	Cellular responses to stimuli and Programmed Cell Death.
Scgb3a2	Secretoglobin Family 3A Member 2	secreted lung surfactant protein
Scgb1a1	Secretoglobin Family 1A Member 1	implicated in numerous functions including anti-inflammation, inhibition of phospholipase A2 and the sequestering of hydrophobic ligands
Gpam	Glycerol-3-Phosphate Acyltransferase, Mitochondrial	acyltransferase activity and glycerol-3-phosphate O-acyltransferase activity
Cdh11	Cadherin 11	integral membrane proteins that mediate calcium-dependent cell-cell adhesion.
Ldlr	Low Density Lipoprotein Receptor	cell surface proteins involved in receptor-mediated endocytosis of specific ligands.
Myocd	Myocardin	transcriptional co-activator of serum response factor (SRF)

Table 4

List of top dysregulated genes on exposure to differently flavored (fruit, menthol and tobacco) e-cig aerosol in C57Bl/6J mouse lungs

GO ID	ONTOLOGY	Description	Gene ID	BgRatio	p.adjust
GO:0045907	BP	positive regulation of vasoconstriction	Htr2c/Chrm3/Add3	53/23062	0.033815
GO:00192	BP	regulation of	Htr2c/Chrm3/Add3	86/2306	0.04249

29		vasoconstriction		2	1
GO:1903978	BP	regulation of microglial cell activation	Stap1/Ldlr	16/23062	0.042491
GO:0002683	BP	negative regulation of immune system process	Hmgb3/Gpam/Scgb1a1/Stap1/Ldlr	464/23062	0.042491
GO:0010867	BP	positive regulation of triglyceride biosynthetic process	Gpam/Ldlr	19/23062	0.042491
GO:0042310	BP	vasoconstriction	Htr2c/Chrm3/Add3	109/23062	0.042491
GO:0046889	BP	positive regulation of lipid biosynthetic process	Htr2c/Gpam/Ldlr	110/23062	0.042491
GO:0010866	BP	regulation of triglyceride biosynthetic process	Gpam/Ldlr	25/23062	0.047046
GO:0001919	BP	regulation of receptor recycling	Ldlr/Ramp3	29/23062	0.047046
GO:0007271	BP	synaptic transmission, cholinergic	Htr2c/Chrm3	29/23062	0.047046
GO:0150077	BP	regulation of neuroinflammatory response	Stap1/Ldlr	29/23062	0.047046
GO:0090208	BP	positive regulation of triglyceride metabolic process	Gpam/Ldlr	31/23062	0.047046
GO:0045987	BP	positive regulation of smooth muscle contraction	Chrm3/Myocd	35/23062	0.047046
GO:0097242	BP	amyloid-beta clearance	Ldlr/Myocd	36/23062	0.047046
GO:0040013	BP	negative regulation of locomotion	Htr2c/Mitf/Stap1/Myocd	360/23062	0.047046
GO:0010667	BP	negative regulation of cardiac muscle cell apoptotic process	Acot1/Myocd	37/23062	0.047046
GO:0001881	BP	receptor recycling	Ldlr/Ramp3	40/23062	0.047046
GO:0010664	BP	negative regulation of striated muscle cell apoptotic process	Acot1/Myocd	40/23062	0.047046
GO:0019432	BP	triglyceride biosynthetic process	Gpam/Ldlr	40/23062	0.047046
GO:0035296	BP	regulation of tube diameter	Htr2c/Chrm3/Add3	173/23062	0.047046
GO:0097746	BP	blood vessel diameter maintenance	Htr2c/Chrm3/Add3	173/23062	0.047046
GO:00017	BP	microglial cell activation	Stap1/Ldlr	42/2306	0.04704
74				2	6
GO:1903725	BP	regulation of phospholipid metabolic process	Htr2c/Ldlr	42/23062	0.047046
GO:0035150	BP	regulation of tube size	Htr2c/Chrm3/Add3	174/23062	0.047046

Table 5

Gene Ontology results showing the top hits from the commonly dysregulated genes in all clusters on exposure to e-cig aerosols

Of note, the data presented in this study is a sub-part of a larger study. In addition to the groups mentioned in this manuscript, we also had two additional groups of Tobacco-Derived (TDN) and Tobacco-Free Nicotine (TFN). Though further objectives and experimentations performed in both these studies were varied, common air and PG:VG samples were used for analyses of cytokine/chemokine and cell count data as described in our recent publication⁽⁵⁰⁾.

Discussion

E-cigs and associated products have constantly been under scrutiny by the US Food and Drug Administration (FDA) due to public health concerns. In February 2020, FDA placed a regulation on all cartridge-based flavored e-cigs except for menthol and tobacco to reduce the use of e-cigs amongst adolescents and young adults. But it left a loophole for the sale of flavored (including menthol) disposable and open system e-cigs^(1, 51). Importantly, most e-cig related bans in the US have happen at the state level, thus allowing differential levels of restrictions imposed on the premarket tobacco applications (PMTAs) and sales which defeats the purpose of limiting their accessibility to the general public^(52, 53). Each year new products are introduced in the market with newer device designs and properties, to lure the users (adults between the ages of 18-24 years) which makes it crucial to continue with the assessments of toxicity and health effects of e-cigs in an unbiased manner⁽⁵⁴⁾.

Numerous studies indicate increased oxidative stress, DNA damage, and loss of neutrophil function due to exposure to e-cig aerosols *in vitro* and *in vivo*^(7, 30, 31, 50, 55, 56). However, we do not have much knowledge about the cell populations and biological signaling mechanisms that are most affected upon exposure to differently flavored e-cig aerosols. To bridge this gap in knowledge, we studied the transcriptional changes in the inflammatory responses due to acute (1-hr nose-only exposure including 120 puffs for 5-consecutive days) exposure to fruit-, menthol- and tobacco-flavored e-cig aerosols using single-cell technology.

Reports indicate that the release of metal ions due to the burning of metal coil is a major source of variation during e-cig exposures^(57–59). A 2021 study reported the presence of 21 elements in the pod atomizers from different manufacturers identifying a high abundance of 11 elements including nickel, iron, zinc, and sodium amongst others⁽⁶⁰⁾. Previous studies have also shown the presence of similar elements in the e-cig aerosols which could have a possible adverse health effect on the vapers^(15, 61). But more importantly, our study points towards a much graver issue, which pertains to the product design of the e-cig vapes. We believe that the aerosol composition varies based on the type of atomizer, coil resistance, coil composition, and chemical reactivity of the e-liquid being used. While much work is done on the chemical composition of the flavors and e-liquids, the other aspects of device design remain understudied and must be an area of research in the future. Another factor that may limit the interpretation of our results pertains to the correlation between the leached metal and the observed transcriptional changes. Since our study provides proof of day-to-day variation in the leaching of metal ions from the same liquid using the same atomizer, it could be possible to develop a statistical model correlating the differential metal exposure to the gene expression changes. We do not conduct such analyses as this was not the focus of this work, but it is a possibility that could be explored in the future.

E-cig vaping has been known to affect the innate and adaptive immune responses among vapers^(30, 62–64), but flavor-specific effects on immune function are not fully explored. While we expected to see flavor-dependent changes in our experiment, we did not anticipate observing interesting sex-dependent variation in the lung tissues at single-cell level. In this respect, recent studies show concurring evidence suggesting sex-specific changes in lung inflammation, mitochondrial damage, gene expression, and even DNA methylation in mice exposed to e-cig aerosols^(65, 66).

One of the most interesting discoveries from our single-cell analyses was the identification of a new cluster of neutrophils that was called Ly6G-neutrophils in this study. While we observed an increase in the cell percentages of these cells in our treatment groups, little to no change in the gene expression was noted in this study which could be indicative of impaired function of these neutrophil population, a fate being reported by various previous studies pertaining to e-cig exposures (13 [↗](#), 62 [↗](#)). In fact, our study identified this unique population and showed that IL-1 mediated responses were dampened in tobacco and menthol-exposed lung neutrophils which was corroborated by mild to no change in the levels of IL-1 β in the lung tissue digests from exposed mice as compared to air. Ly6G is an important marker of neutrophil maturation in mammalian cells and has been reported in relation to various bacterial and parasitic infections in previous studies (67 [↗](#), 68 [↗](#)). Exposure to e-cig aerosols may result in a loss of function of mature neutrophils in the mouse lungs which gets actively replaced by inactive/immature neutrophils that are deficient in Ly6G expression. In fact, this could explain our inability to identify significant changes in mature neutrophil population through flow cytometry (where Ly6G was used as a marker) when comparing treated and control groups. Interestingly, our *in vivo* findings were corroborated by a recently published work where neutrophils from healthy volunteers demonstrated a reduction in neutrophil chemotaxis, phagocytic function, and neutrophil extracellular trap formation on exposure to e-cig aerosols, thus suggesting loss of function by mature neutrophils (30 [↗](#)). However, this study did not investigate the effects on the immature neutrophil population, an area that needs further study in future research.

Of note, myeloid and lymphoid limbs of immunity are interconnected (69 [↗](#), 70 [↗](#)). A decline in the neutrophil-mediated immunity might activate other cell types to offer protection. In our study, we find a decline in the neutrophilic immune responses in menthol and tobacco flavored e-cig exposed mouse lungs. Contrarily, we report an increase in the T-cell responses in the form of increased CD4⁺ and CD8⁺ T cells from both scRNA seq and flow cytometric analyses. In fact, increased expression of genes including *Malt1*, *Serpinb9b*, and *Sema4c* are indicative of enhanced T-cell mediated immune response in the lungs of mice exposed to fruit(mango) flavored e-cig aerosols (71 [↗](#)–73 [↗](#)). Contrary to this, exposure to menthol-flavored e-cig aerosols had a much milder effect on the lymphoid population within the lungs of C57Bl/6J mice. We found evidence for increased cell proliferation due to activation of cyclin-dependent protein kinase signaling mediated via expression of genes including *Cdk8* and *Camk1d* in these cells (42 [↗](#), 74 [↗](#)). Exposure to tobacco-flavored e-cig aerosol provided evidence for decreased chaperone-mediated protein folding, due to the downregulation of Chaperonin Containing TCP-1 (CCT) family of proteins. Chaperonin Containing TCP-1 proteins are important to regulate the production of native active, tubulin, and other proteins crucial for cell cycle progression and cytoskeletal organization (75 [↗](#)). This is in conjunction with the upregulation of *Klra4* and *Klra8* that is indicative of increased protein misfolding and cytotoxic responses in the lymphoid cells of tobacco-exposed e-cig aerosols (76 [↗](#), 77 [↗](#)).

Overall, we provide evidence of subdued innate immune responses due to loss of function of neutrophils and increased T-cell proliferation and cytotoxicity in a flavor-dependent manner upon exposure to e-cig aerosols in this study. An increase in the levels of CCL17, CCL20, CCL22, IL2 and Eotaxin in the lung digests from tobacco-exposed mouse lungs further support this deduction as these cytokines/chemokines are associated with T-cell mediated immune responses (78 [↗](#)–85 [↗](#)). Importantly CXCL16 attracts T-cells and natural killer cells to activate cell death. It has been reported to be involved in LPS-mediated acute lung injury (ALI), an outcome which has also been linked with e-cig exposures in human (86 [↗](#), 87 [↗](#)).

In the end, we compiled a list of commonly dysregulated genes in a flavor independent manner and identified 29 gene targets. Signal-transducing adaptor protein-2 (*Stap1*) which is commonly upregulated upon exposure to e-cig aerosols is known to regulate T-cell activation and airway inflammation which falls in line with the overall crux of our findings (88 [↗](#)). Another gene that was found to be consistently dysregulated in many cell types was Cold-inducible RNA binding

protein (*Cirbp*). *Cirbp* is a stress response proteins linked with stressors like hypoxia. Its upregulation upon e-cig exposure supports that vaping induces oxidative stress and can have adverse implications on the exposed cell types (89). 5-hydroxytryptamine receptor 2C (*Htr2c*) and *Klra8* are other genes in this category of commonly dysregulated genes that are associated with enhancing inflammation and cell death (76, 77, 90). Overall, we provide a cell-specific atlas of immune responses upon exposure to differently flavored e-cig aerosols.

Considering that scRNA technology has not been commonly used for e-cig research, ours is one of the first studies employing this technique to identify possible changes in the cellular composition and gene expressions. Importantly we use the nose-only exposure system for our experiment to avoid exposure through other routes. However, despite the novel approach and state-of-the-art exposure system, we had a few limitations. First, we used a small sample size to identify the changes in the mouse lungs upon exposure to e-cig aerosols at a single-cell level. Due to the expensive nature of single-cell sequencing technology and limited information in the literature reporting changes at single-cell level, we chose to design this experiment with small sizes of experimental and validation cohort. But, based on the encouraging findings from this pilot study, future studies could be designed with a larger sample size and longer durations of exposure to identify the acute and chronic effects of vaping. Second, we could not delineate the sex-dependent changes upon exposure from our data. While we found evidence of sex-dependent differences in immune responses on exposure to differently flavored e-cig aerosols we pooled the samples from 2 mice/sex/group to perform scRNA seq analyses. This limits us from analyzing and drawing conclusions with regards to the sex-dependent variations at the level of differential gene expression in our data with confidence. Thus, we chose to use aggregate analyses to make comparisons. However, future experimental designs must study sex as a confounder for studying the effects of e-cig exposure in humans. Third, the inclusion of PGVG+Nic group was streamlined in this study, but future work including this group for scRNA seq analyses might be crucial to delineate the effects of nicotine alone on gene transcription. Fourth, we did not anticipate changes in the metal release on consecutive days of exposure in our study. Our data pointed towards the importance of device design in e-cig exposures. Future studies need to identify the factors that may affect the daily composition of e-cig aerosols and devise a method of better monitoring these possible confounders. However, in this regard our experiment does mimic the real-life well, as such variations due to prolonged storage of e-liquid and differences arising due to vape design must be common amongst human vapers.

In conclusion, we identified cell-specific changes in the gene expressions upon exposure to e-cig aerosols using single cell technology. We identified a set of top 29 dysregulated genes that could be studied as markers of toxicity/immune dysfunction in e-cig research. Future work with larger sample sizes and sex-distribution is warranted to understand the health impacts of long-term use of these novel products in humans.

Methods

Animals Ethics Statement

All experiments were conducted per the guidelines set by the University Committee on Animal Resources at the University of Rochester Medical Center (URMC). Care was taken to implement an unbiased and robust approach during the experimental design and conduction of each experiment to ensure data reproducibility per the National Institutes of Health standards.

Animals

We ordered 5-week-old pups of male and female C57BL/6J mice from Jackson Laboratory to conduct this experiment. Prior to the start of the experiment, mice were housed at the URMC Vivarium for acclimatization. Thereafter the animals were moved to the mouse Inhalation Facility

at URMIC for training and exposures.

One week prior to the start of the exposures, mice underwent a five-day nose-only training to adapt themselves to the mesh restraints of the exposure tower. The mouse restraint durations were increased gradually to minimize the animal stress and discomfort. Of note, the mouse sacrifice was performed within 8-12 weeks' age for each mouse group to ensure that the mouse age corresponds to the age of adolescents (12-17 years) in humans (91 [↗](#), 92 [↗](#)).

E-cigarette Device and E-liquid

We utilized an eVic-VTC mini and CUBIS pro atomizer (SCIREQ, Montreal, Canada) with a BF SS316 1.0-ohm coil from Joyetech (Joyetech, Shenzhen, China) for vaping and the inExpose nose-only inhalation system from SCIREQ (SCIREQ, Montreal, Canada) for mouse exposures. Both air and PG:VG exposed mice groups were considered as controls for this experiment. We used commercially available propylene glycol (PG; EC Blend) and vegetable glycerin (VG; EC Blend) in equal volume to prepare a 50:50 solution of PG:VG. For flavored product exposures, mice were exposed to three different e-liquids – a menthol flavor “Menthol-Mint”, a fruit flavor “Mango” and a tobacco flavor “Cuban Blend”. Of note, all the e-liquids were commercially manufactured with 50mg/mL of tobacco derived nicotine (TDN). So, all treatments have nicotine in addition to the flavoring mentioned respectively. Additionally, we used a mixture of PG:VG with 50mg/mL of TDN as a control for limited experiments to study the effect of nicotine alone in our treatment. This group is labeled as PGVG+Nic for the rest of the manuscript.

E-cigarette Exposure

Scireq Flexiware software with the InExpose Inhalation system was used for controlling the Joyetech eVic-VTC mini device to perform nose-only mouse exposures. For this exposure, we utilized a puffing profile that mimicked the puffing topography of e-cig users in two puffs per minute with a puff volume of 51 mL, puff duration of 3 seconds, and an inter puff interval of 27 seconds with a 2 L/min bias flow between puffs (93 [↗](#)). **Figures 1A** [↗](#) and **1B** [↗](#) depict the experimental design and exposure system employed for this study.

Age-matched male and female (n=5 per sex) mice were used for each group, namely, air, PG:VG, Fruit, Menthol, and Tobacco. To ensure rigor and reproducibility in our work, we have used age- and sex-matched control and treated mice in this study. Confounders like environment and stress were minimized by housing all the cages in environmentally controlled conditions and training all the mice (both control and treated) in nose-only chambers. Each group of mice was exposed to the above-mentioned puffing profile for one hour each day (120 puffs) for a total of five consecutive days. Additionally, a group of mice (n=3/sex) was exposed to PGVG+Nic for the same duration using similar exposure profile to serve as control to assess the effect of nicotine on the observed changes using selected experiments. Air-exposed mice were exposed to the same puffing profile for a total of five consecutive days to ambient air. We recorded the temperature, humidity, and CO levels of the aerosols generated at the start, mid, and end of the exposure on each day using the Q-Trak Indoor Air Quality Monitor 7575 (TSI, Shoreview, MN). Total Particulate Matter (TPM) sampling was done from the exhaust tubing of the set-up at the 30 min mark of the exposure and at the inlet connected to the nose-only tower (shown in **Figure 1B** [↗](#)) immediately after the culmination of the exposure. Gravimetric measurements for TPM were also conducted to confirm relative dosage to each mouse group daily.

Preparation of single-cell suspension

The animals were sacrificed (at 8-10 weeks' age) immediately after the final exposure. Vascular lung perfusion was performed using 3mL of saline before harvesting the lung lobes for preparation of single-cell suspension. It is important to mention here that of the 5 lung samples/sex/group; 2 sex/group were used for histological assessments and scRNA analyses. Here,

the left lung lobe was inflated using low-melting agarose and used for histology, while the rest of the uninflated lung lobes were used for preparing the single-cell suspension. We pooled the lung lobes for each sex per group for preparation of the single-cell suspension as depicted in **Figure 1A**. The lung lobes were weighed digested using Liberase method as described earlier(94).

Library Preparation and Single-Cell Sequencing

The prepared single-cell suspension was sent to the Genomics Research Center (GRC) at URM for library preparation and single-cell sequencing. Library preparation was performed from control and treatment groups using the 10X single-cell sequencing pipeline by 10X Genomics and 10000 cells were captured per sample using the Chromium platform. The prepared library was sequenced on NovaSeq 6000 (Illumina, San Diego, CA) at a mean sequence depth of 30,000 reads per cell. Read alignment was performed to GRCh38 Sequence.

Data Analyses

We used the standard Seurat v4.2 analyses pipeline to analyze our data. In brief the low-quality cells and potential doublets were excluded from the dataset to create the analyses dataset. The residual features due to the presence of RBCs were corrected before integration. “scTransform” function was used for integration of all the datasets after which the standard Seurat pipeline was used for data normalization of integrated data. “FindVariableGenes” gene function was used to identify the variable genes for dimensionality reduction using PCA function. UMAP was used for dimensionality reduction and clustering of the cells.

To identify the unique features and cell clusters within each cell subtypes, we used the sub-setting feature within Seurat. After identifying the 5 major cell populations (epithelial, endothelial, stromal, myeloid and lymphoid) in our data sets, each of these cell types were sub-clustered using “subset ()” function, normalized and re-clustered. Cell annotation for each of the subsets was performed with the help of Tabula Muris database. However, some clusters were annotated manually with the help of a literature search as discussed in the results section.

DESeq2 was used to perform pseudobulk analyses to identify differentially expressed genes within each group. The ClusterProfiler was employed to perform gene enrichment analyses of the differentially expressed genes.

Cytokine/chemokine assessment

We used multiplex assay to determine the levels of cytokine/chemokine in the lung homogenates from control and e-cig aerosol exposed mouse lungs using commercially available Bio-Plex Pro Mouse Chemokine Assay (Cat#12009159, Bio-RAD, Hercules, CA) per the manufacturer’s instructions. Approximately 40 mg of mouse lung lobes were homogenized in 300uL of 1X RIPA buffer with 0.1% protease and phosphatase inhibitor. The lung homogenate was stored on ice for 30 min. Following incubation, the homogenate was centrifuged at 15000 rpm for 15 min at 4 degrees Celsius. The supernatant was collected and used for performing the multianalyte assay for determination of cytokine/chemokine levels using Luminex FlexMap3D system

Lung Histology

The left lung lobe of mice used for scRNA seq were inflated with 1% low melting agarose and fixed with 4% neutral buffered PFA. Fixed lungs were dehydrated, before being paraffin-embedded and sectioned (5µm). Hematoxylin and eosin (H&E) staining was performed by the Histology, Biochemistry, and Molecular Imaging Core at URM. We employed MetaMorph software (Molecular Devices) to determine the “Mean linear intercept” (Lm) of airspace. In brief, 6-10 randomly selected ×100 fields per slide were photographed in a blinded manner, and Lm values

were quantitated for each captured image using the MetaMorph software. Care was taken to avoid pictures with bronchiolar regions and knife marks in the tissue for quantitation. An average of all quantified images was then plotted for each sample in a sex-dependent manner.

Flow cytometry

Flow cytometry was performed on the cells collected from BALF and lung homogenates from air and flavored e-cig aerosol exposed mouse lungs. For analyses of cells collected from BALF, the fluid was centrifuged at 2000g for 10 min at 4°C. Supernatant was collected for ELISA, while the cell pellet was re-suspended in 1mL of PBS. For analyses of immune cell population in the lung, the lung lobes were digested as described earlier (94). The single-cell suspension thus prepared was used to run flow cytometry using the BD LSRFortessa cell analyzer. Cells were blocked with CD16/32 (Tonbo biosciences 70-0161-u500, 1:10) to prevent nonspecific binding and stained with a master mix of Siglec F (BD OptiBuild Cat#740280, 1:200), CD11b (Biolegend Cat #101243, 1:200), Ly6G (BD Horizon Cat# 562700, 1:200), CD45 (Biolegend Cat#103126, 1:200), CD11c (Biolegend Cat #117318, 1:200), CD4 (Biolegend Cat#116012, 1:200), and CD8 (eBiosciences Cat#17-0081-82, 1:200). 7AAD (eBiosciences Cat#00-6993-50, 1:10) was used as the nucleic acid dye to detect live and dead cells.

Metal analyses

To understand the levels of metals released during subsequent days of exposure, we performed Inductively coupled plasma mass spectrometry (ICP-MS) on the e-cig aerosol condensates collected from each day of exposure using a Perkin Elmer ICP-MS model 2000C. The samples were run using a Total Quant KED protocol with 4mL/min Helium flow and externally calibrated using a blank and a 100ppb standard for the 51 elements. The samples were submitted to the Element Analyses facility at URM, and levels of metals thus detected were plotted.

Statistical significance

We used GraphPad Prism 10.0 for all statistical calculations. All the data plotted in this paper are expressed as mean $\pm$ SEM. Pairwise comparisons were done using unpaired *t* test while one-way analysis of variance (ANOVA) with ad-hoc Tukey's test was employed for multi-group comparisons. To identify sex-based variations in our treatment groups, Tukey post hoc two-way ANOVA was employed.

Data availability

The datasets generated during and/or analyzed during the current study are deposited on NCBI Gene Expression Omnibus. Specifically, mouse scRNAseq is available under accession code "GSE263903" (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE263903>). The data will be publicly available upon publication or on April 11, 2025, whichever is earlier. Reviewer's token for access is 'wt0fqmayjdeltcn'. All other relevant data supporting the key findings of this study are available within the article and its Supplementary Information files or from the corresponding author upon reasonable request. Source data are provided with this paper.

Code Availability

Data collection was performed with mkfastq pipeline in Cell Ranger's (v7.0.1). Cell Ranger (v7.0.1) was used for cell and gene counting using the default settings. Single-cell analysis was performed using the Seurat R package (v4.3.0) using the recommended workflow.

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Additional information

Author's contributions

GK, TL and AT planned and conducted the experiments; TL analyzed the lab-based assays; GK and AT analyzed the scRNA data; GK wrote and edited the manuscript; TL, AT and IR reviewed the data and manuscript; IR conceived and procured funding for the project.

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Reviewer #1 (Public review):

Summary:

The authors tackled the public concern about E-cigarettes among young adults by examining the lung immune environment in mice using single-cell RNA sequencing, discovering a subset of Ly6G⁺ neutrophils with reduced IL-1 activity and increased CD8 T cells following exposure to tobacco-flavored e-cigarettes. Preliminary serum cotinine (nicotine metabolite) measurements validated the effective exposure to fruit, menthol, and tobacco-flavored e-cigarettes with air and PG:VG serving as control groups. They also highlighted the significance of metal leaching, which fluctuated over different exposure durations to flavored e-cigarettes, underscoring the inherent risks posed by these products. The scRNAseq analysis of e-cig exposure to flavors and tobacco demonstrated the most notable differences in the myeloid and lymphoid immune cell populations. Differentially expressed genes (DEGs) were identified for each group and compared against the air control. Further sub-clustering revealed a flavor-specific rise in Ly6G⁺ neutrophils and heightened activation of cytotoxic T cells in response to tobacco-flavored e-cigarettes. These effects varied by sex, indicating that immune changes linked to e-cig use are dependent on gender. By analyzing the expression of various genes and employing gene ontology and gene enrichment analysis, they identified key pathways involved in this immune dysregulation resulting from flavor exposure. Overall, this study affirmed that e-cigarette exposure can suppress the neutrophil-mediated immune response, subsequently enhancing T cell toxicity in the lung tissue of mice.

Strengths:

This study used single-cell RNA sequencing to comprehensively analyze the impact of e-cigarettes on the lung. The study pinpointed alterations in immune cell populations and identified differentially expressed genes and pathways that are disrupted following e-cigarette exposure. The manuscript is well written, the hypothesis is clear, the experiments are logically designed with proper control groups, and the data is thoroughly analyzed and presented in an easily interpretable manner. Overall, this study suggested novel mechanisms by which e-cigs impact lung immunity and created a dataset that could benefit the lung immunity field.

Weaknesses:

(1) The authors included a valuable control group - the PG:VG group, since PG:VG is the foundation of the e-liquid formulation. However, most of the comparative analyses use the air group as the control. Further analysis comparing the air group to the PG:VG group, and the PG:VG group to the individual flavored e-cig groups will provide more clear insights into the true source of irritation. This is done for a few analyses but not consistently throughout the paper. Flavor-specific effects should be discussed in greater detail. For example, Figure 1E shows that the Fruit flavor group exhibits more severe histological pathology but similar effects were not corroborated by the single-cell data.

(2) The characterization of Ly6g⁺ vs Ly6g⁻ neutrophils is interesting and potentially very impactful. Key results like this from scRNAseq analyses should be validated by qPCR and flow cytometry.

Also, a recent study by Ruscitti et al reported Ly6g⁺ macrophages in the lung which can potentially confound the cell type analysis. A more detailed marker gene and sub-population analysis of the myeloid clusters could rule out this potential confounding factor.

<https://doi.org/10.7554/eLife.106380.1.sa3>

Reviewer #2 (Public review):

This study provides some interesting observations on how different flavors of e-cigarettes can affect lung immunology, however there are numerous flaws including a low number of replicates and a lack of effective validation methods which reduces the robustness and rigor of the findings.

Strengths:

The strength of the study is the successful scRNA-seq experiment which gives good preliminary data that can be used to create new hypotheses in this area.

Weaknesses:

The major weakness is the low number of replicates and the limited analysis methods. Two biological n per group is not acceptable to base any solid conclusions. Any validity data was too little (only cell % data) and did not always support the findings (e.g. Figure 4D does not match 4C). Often n seems to be combined and only one data point is shown, it is not at all clear how the groups were analysed and how many cells in each group were compared.

Other specific weaknesses were identified in addition to the ones above:

- (1) Only 71,725 cells means only 7,172 per group, which is 3,586 per animal - how many of these were neutrophils, T-cells, and macrophages? This was not shown and could be too low.
- (2) The dynamic range of RNA measurement using scRNAseq is known to be limited - how do we know whether genes are not expressed or just didn't hit detection? This links into the Ly6G negative neutrophil comment, but in general, the lack of gene expression in this kind of data should be viewed with caution, especially with a low n number and few cells.
- (3) There is no rigorous quantification of Ly6G⁺ and Ly6G⁻ cells in the flow cytometry data.
- (4) Eosinophils are heavily involved in lung biology but are missing from the analysis.
- (5) The figures had no titles so were difficult to navigate.
- (6) PGVG is not defined and not introduced early enough.
- (7) Neutrophils are not well known to proliferate, so any claims about proliferation need to be accompanied by validation such as BrdU or other proliferation assays.
- (8) It was not clear how statistics were chosen and why Table S2 had a good comparison (two-way ANOVA with gender as a variable) but this was not used for other data particularly when looking at more functional RNA markers (Table S2 also lacks the interaction statistic which is most useful here).

(9) Many statistics are only vs air control, but it would be more useful as a flavour comparison to see these vs PGVG. In some cases, the carrier PGVG looks worse than some of the flavours (which have nicotine).

(10) The n number is a large issue, but in Figures such as 4, 6, and 7 it could be a bigger factor. The number of significant genes identified has been determined by chance rather than any real difference, e.g. Is Il1b not identified in Fruit flavour vs air because there wasn't enough n, while in Air vs Tobacco, it randomly hit the significance mark. This is but an example of the problems with the analysis and conclusions

(11) The data in Figure 7A is confusing, if this is a comparison to air, then why does air vs air not equal 1? Even if this was the comparison to the average of air between males and females, then this doesn't explain why CCL12 is >1 in both. Is this z-score instead? Regardless the data is difficult to interpret in this format.

(12) Individual n was not shown for almost all experiments - e.g. Figure 1D - what is this representative of? Figure 2D - is this bulk-grouped data for all cells and all mice? The heatmaps are also pooled from 2n and don't show the variability.

<https://doi.org/10.7554/eLife.106380.1.sa2>

Reviewer #3 (Public review):

This work aims to establish cell-type specific changes in gene expression upon exposure to different flavors of commercial e-cigarette aerosols compared to control or vehicle. Kaur et al. conclude that immune cells are most affected, with the greatest dysregulation found in myeloid cells exposed to tobacco-flavored e-cigs and lymphoid cells exposed to fruit-flavored e-cigs. The up-and-down-regulated genes are heavily associated with innate immune response. The authors suggest that a Ly6G-deficient subset of neutrophils is found to be increased in abundance for the treatment groups, while gene expression remains consistent, which could indicate impaired function. Increased expression of CD4+ and CD8+ T cells along with their associated markers for proliferation and cytotoxicity is thought to be a result of activation following this decline in neutrophil-mediated immune response.

Strengths:

(1) Single-cell sequencing data can be very valuable in identifying potential health risks and clinical pathologies of lung conditions associated with e-cigarettes considering they are still relatively new.

(2) Not many studies have been performed on cell-type specific differential gene expression following exposure to e-cig aerosols.

(3) The assays performed address several factors of e-cig exposure such as metal concentration in the liquid and condensate, coil composition, cotinine/nicotine levels in serum and the product itself, cell types affected, which genes are up- or down-regulated and what pathways they control.

(4) Considerations were made to ensure clinical relevance such as selecting mice whose ages corresponded with human adolescents so that the data collected was relevant.

Weaknesses:

(1) The exposure period of 1 hour a day for 5 days is not representative of chronic use and this time point may be too short to see a full response in all cell types. The experimental design is not well-supported based on the literature available for similar mouse models.

(2) Several claims lack supporting evidence or use data that is not statistically significant. In particular, there were no statistical analyses to compare results across sex, so conclusions stating there is a sex bias for things like Ly6G⁺ neutrophil percentage by condition are observational.

(3) Statistical analyses lack rigor and are not always displayed with the most appropriate graphical representation.

(4) Overall, the paper and its discussion are relatively limited and do not delve into the significance of the findings or how they fit into the bigger picture of the field.

(5) The manuscript lacks validation of findings in tissue by other methods such as staining.

(6) This paper provides a foundation for follow-up experiments that take a closer look at the effects of e-cig exposure on innate immunity. There is still room to elaborate on the differential gene expression within and between various cell types.

<https://doi.org/10.7554/eLife.106380.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors tackled the public concern about E-cigarettes among young adults by examining the lung immune environment in mice using single-cell RNA sequencing, discovering a subset of Ly6G⁺ neutrophils with reduced IL-1 activity and increased CD8 T cells following exposure to tobacco-flavored e-cigarettes. Preliminary serum cotinine (nicotine metabolite) measurements validated the effective exposure to fruit, menthol, and tobacco-flavored e-cigarettes with air and PG/VG serving as control groups. They also highlighted the significance of metal leaching, which fluctuated over different exposure durations to flavored e-cigarettes, underscoring the inherent risks posed by these products. The scRNAseq analysis of e-cig exposure to flavors and tobacco demonstrated the most notable differences in the myeloid and lymphoid immune cell populations. Differentially expressed genes (DEGs) were identified for each group and compared against the air control. Further sub-clustering revealed a flavor-specific rise in Ly6G⁺ neutrophils and heightened activation of cytotoxic T cells in response to tobacco-flavored e-cigarettes. These effects varied by sex, indicating that immune changes linked to e-cig use are dependent on gender. By analyzing the expression of various genes and employing gene ontology and gene enrichment analysis, they identified key pathways involved in this immune dysregulation resulting from flavor exposure. Overall, this study affirmed that e-cigarette exposure can suppress the neutrophil-mediated immune response, subsequently enhancing T cell toxicity in the lung tissue of mice.

Strengths:

This study used single-cell RNA sequencing to comprehensively analyze the impact of e-cigarettes on the lung. The study pinpointed alterations in immune cell populations and identified differentially expressed genes and pathways that are disrupted following e-cigarette exposure. The manuscript is well written, the hypothesis is clear, the experiments are logically designed with proper control groups, and the data is thoroughly analyzed and presented in an easily interpretable manner. Overall, this study

suggested novel mechanisms by which e-cigs impact lung immunity and created a dataset that could benefit the lung immunity field.

We thank the reviewer for identifying the strengths of our work.

Weaknesses:

The authors included a valuable control group - the PG/VG group, since PG/VG is the foundation of the e-liquid formulation. However, most of the comparative analyses use the air group as the control. Further analysis comparing the air group to the PG/VG group, and the PG/VG group to the individual flavored e-cig groups will provide more clear insights into the true source of irritation. This is done for a few analyses but not consistently throughout the paper. Flavor-specific effects should be discussed in greater detail. For example, Figure 1E shows that the Fruit flavor group exhibits more severe histological pathology, but similar effects were not corroborated by the single-cell data.

We thank the reviewer for this query. We agree that PG/VG group is the foundation of the e-liquid formulation and hence comparisons with this group is of significance to understand the effect of individual flavors on the cell population. Though we compared the flavored e-cig groups with PG/VG group, we did not discuss it in detail within the manuscript to avoid confusions in interpretation for such a big dataset. However, we will include the comparisons with the PG/VG group as a Supplement File in our revised manuscript to facilitate proper interpretation of our omics data to interested readers.

While we agree that flavor-specific effects might be of interest, we did not delve into exploring them in detail as the fruit flavored e-liquids have now been regulated for sale in the US. Thus, from regulatory point of view, the effects of tobacco- and menthol-flavored e-liquids hold most interest. Since at the time of conducting this study, fruit flavors were in the market, we have still included the data. However, studying it further was not the focus of this work. Nevertheless, interested readers of our manuscript can have access to our dataset to allow further analyses and interpretation of our results.

The characterization of Ly6g+ vs Ly6g- neutrophils is interesting and potentially very impactful. Key results like this from scRNAseq analyses should be validated by qPCR and flow cytometry.

Also, a recent study by Ruscitti et al reported Ly6g+ macrophages in the lung which can potentially confound the cell type analysis. A more detailed marker gene and sub-population analysis of the myeloid clusters could rule out this potential confounding factor.

We agree with the reviewer that the loss of Ly6G on neutrophils is a very interesting find and we are in process of designing neutrophil specific experiments to study the impact of e-cig exposure on neutrophil maturation and function which will be discussed in subsequent work by our group. However, to address the concerns raised by the reviewer, we are staining the lung tissue samples from air-and differently flavored e-cig aerosol exposed mouse lungs with Ly6G and S100A8 (universal marker for neutrophil) to see the infiltration of Ly6g+ vs Ly6g- neutrophils within the lungs of exposed and unexposed mice. This would also address the question if these populations were neutrophils or belong to another myeloid origin as suggested by recent publications. We will share the results from our findings in the revised manuscript and update our interpretations accordingly with better validations.

Reviewer #2 (Public review):

This study provides some interesting observations on how different flavors of e-cigarettes can affect lung immunology, however there are numerous flaws including a low number of replicates and a lack of effective validation methods which reduces the robustness and rigor of the findings.

Strengths:

The strength of the study is the successful scRNA-seq experiment which gives good preliminary data that can be used to create new hypotheses in this area.

We appreciate the reviewer for recognizing the strength of this work.

Weaknesses:

The major weakness is the low number of replicates and the limited analysis methods. Two biological n per group is not acceptable to base any solid conclusions. Any validity data was too little (only cell % data) and did not always support the findings (e.g. Figure 4D does not match 4C). Often n seems to be combined and only one data point is shown, it is not at all clear how the groups were analyzed and how many cells in each group were compared.

We thank the reviewer for the critique to allow us to improve our analyses. We understand that the low number of replicates in this work makes the analyses difficult to draw solid conclusions, but this was a pilot study to understand the changes in the mouse lung upon acute exposures to flavored e-cig aerosols at a single cell level. So far, the e-cig field has been primarily focused on conducting toxicological studies to help regulatory bodies to set standards and enforce laws to better regulate the manufacture, sale and distribution of e-cig products. However, adolescents and young adults are still getting access to these products, and there is little to no understanding of how this may affect the lung health upon acute and chronic exposures. Single cell technology is a powerful tool to analyze the gene expression changes within cell populations to study cell heterogeneity and function. Yet, it is a costly tool, owing to which, conducting such analyses on large sample sizes is not ideal. This pilot study was designed to get some initial leads for future studies involving larger sample sizes and chronic exposures. Further, we still intend to share our results with the scientific community due to the value of such a dataset for a wider audience interested in learning about the mechanistic underpinnings of e-cig exposures in vivo.

We understand that the validations are limited in our current work and so we are in process of conducting some immunostaining to validate a few targets made through this work. We also want to add here that validating single cell findings using any of the classical methods of experimentation including ELISA, qPCR or flow cytometry is sometimes difficult as many of these techniques still investigate the tissue while the changes shown in single cell analyses are mainly pertaining to a single cell type. This could be a probable reason for the scRNA seq results not aligning with our findings from flow cytometry. The data/findings from this pilot study have now allowed us to be better informed to design an effective flow panel for our future studies. In terms of the statistics and the number of cells for each analysis, we will share the detailed account and information for each to allow better interpretation of our results.

Only 71,725 cells means only 7,172 per group, which is 3,586 per animal - how many of these were neutrophils, T-cells, and macrophages? This was not shown and could be too low.

We do agree that the number of cells could be too low, but to avoid this we never studied the gene expression variations at the finest level of cell identity. We classified the cell clusters into general annotations -myeloid, lymphoid, endothelial, stromal and epithelial- and identified the changes in the gene expressions. Of these, only two clusters (myeloid and lymphoid) with more than ~1000 cells per cell type per group were studied in detail. We will include the cell count information to allow better interpretation of our results in the revised manuscript.

The dynamic range of RNA measurement using scRNAseq is known to be limited - how do we know whether genes are not expressed or just didn't hit detection? This links into the Ly6G negative neutrophil comment, but in general, the lack of gene expression in this kind of data should be viewed with caution, especially with a low n number and few cells.

This is a well-made point, and we thank the reviewer for this comment. We agree that the dynamic range RNA measurement is limited and for low cell numbers that could lead to bias. We are in process of validating the findings regarding the presence of Ly6G⁺ and Ly6G⁻ cells in our control and treated lungs, the outcome of which will be discussed in the revised manuscript. We will also provide the cell number for the Ly6G⁻ cell cluster for each sample with more detailed discussion of our findings. Due to the small sample size and cell capture, few limitations are hard to overcome which will be further elaborated upon in our revisions.

There is no rigorous quantification of Ly6G⁺ and Ly6G⁻ cells in the flow cytometry data.

We understand that flow-based quantification of our scRNA seq findings would be interesting. However, flow cytometry and single cell suspension to perform sequencing were performed parallelly for this study. We used a basic flow panel using single markers to identify individual immune cell type. We did identify changes in the Ly6G population in our treated and control samples using scRNA seq and intend to include it as a marker for our future studies using flow cytometry. But unfortunately, the same analyses could not be performed for the current batch of samples. We will still include results from IHC staining to identify the Ly6G⁺ and Ly6G⁻ population in the lung tissues from control and treated mice in revised manuscript to address some of the concerns raised here.

Eosinophils are heavily involved in lung biology but are missing from the analysis.

We used RBC lysis buffer to remove the excess RBCs during lung digestion for preparation of single cell suspension for scRNA seq in this study. Reports suggest that RBC lysis could adversely affect the eosinophil number and function. We did not identify any cell cluster, representing markers for eosinophils through our scRNA seq data and we believe that our lung digestion protocol could be the reason for the same. We have studied the eosinophil number changes through flow cytometry in these samples and have found significant changes as well. However due to our inability to find cell clusters for eosinophil through scRNA seq data, we did not include these results in the final manuscript. To avoid confusions and maintain transparency we will include our results from flow cytometry experiments in the revised manuscript.

The figures had no titles so were difficult to navigate.

We will make necessary adjustments to the data representation and include the titles to enable easy navigation of the Figures.

PG/VG is not defined and not introduced early enough.

We agree that PG/VG is an important control to compare in e-cig studies. This was the reason why this group was included, and we performed comparisons with this group for scRNA seq studies as well. However, to reduce the complexity of the study, we only shared the comparisons with Air control in this manuscript. We will include the comparisons made with PG/VG group as a Supplementary File in the revised manuscript to allow the interested readers have access to the study results and make necessary interpretations for future research.

Neutrophils are not well known to proliferate, so any claims about proliferation need to be accompanied by validation such as BrdU or other proliferation assays.

We thank the reviewer for this suggestion; however, we cannot perform the BrDU or other proliferation assay on neutrophils for now. We are planning to include these in the study designs of our future work, however we have limitations of funds to continue further experimentation to support this claim for this study. We mention clearly that this is only a scRNA seq finding and requires further study to avoid over-interpretation of our results.

It was not clear how statistics were chosen and why Table S2 had a good comparison (two-way ANOVA with gender as a variable) but this was not used for other data particularly when looking at more functional RNA markers (Table S2 also lacks the interaction statistic which is most useful here).

We thank the reviewer for bringing this concern. We understand that this is a valid point and will include all the necessary information regarding the statistics and other related parameters in the revised manuscript.

Many statistics are only vs air control, but it would be more useful as a flavor comparison to see these vs PG/VG. In some cases, the carrier PG/VG looks worse than some of the flavors (which have nicotine).

We will include the comparisons with PG/VG as supplementary file in our revised manuscript, however we do not intend to describe all those changes in detail in the main manuscript.

The n number is a large issue, but in Figures such as 4, 6, and 7 it could be a bigger factor. The number of significant genes identified has been determined by chance rather than any real difference, e.g. Is Il1b not identified in Fruit flavor vs air because there wasn't enough n, while in Air vs Tobacco, it randomly hit the significance mark. This is but an example of the problems with the analysis and conclusions.

While we agree in part with the concern raised here, we wish to point out that there are limitations to every experiment. In our opinion, an omics study is not necessarily aimed to find the changes at transcript level with absolute certainty, rather to identify probable cell and gene targets to validate with subsequent work. We never claim that our findings are absolute outcomes but rather add the limitation of sample number and need for further research at every step. The strength of this work is to be the first study of its kind looking at changes in the lung cell population at single cell level upon e-cig aerosol exposure. This study has provided us with interesting gene and cell targets that we are now validating with future work. We still strongly believe that a dataset like this is a useful resource for a wider audience to allow efficient study designs and hence it is befitting to be published and discussed amongst our peers.

The data in Figure 7A is confusing, if this is a comparison to air, then why does air vs air not equal 1? Even if this was the comparison to the average of air between males and females, then this doesn't explain why CCL12 is >1 in both. Is this z-score instead? Regardless the data is difficult to interpret in this format.

We thank the reviewer for pointing this out. We realize that the data might be difficult to understand due to scaling of the color codes for the heatmap. We will change the graphical representation and include actual number for fold change in our revised manuscript to allow easy interpretation of these results.

Individual n was not shown for almost all experiments - e.g. Figure 1D - what is this representative of? Figure 2D - is this bulk-grouped data for all cells and all mice? The heatmaps are also pooled from 2n and don't show the variability.

While we have included a pictorial representation of the n number in Figure 1A and mentioned n number in the Figure legends for each figure, we understand that it maybe difficult to navigate. We will attempt to address this in a better manner in the revised manuscript.

However, with respect to the second comment we would like to differ from the reviewer's opinion. Each scRNA seq data had 2 samples – one for male and another for female which has been clearly shown in the current figures. The pooling of cells as mentioned in the comment happened at the stage of preparation of cell suspension from each sex/group at the start of the sequencing. We do not have any means to show the variability amongst pooled samples, which we acknowledge as a shortcoming of our work. So, in terms of representation of the heatmaps and data analyses we have included all the needed information to uphold transparency of our study design and data visualization for each figure and would like to stick to the current representations.

Reviewer #3 (Public review):

This work aims to establish cell-type specific changes in gene expression upon exposure to different flavors of commercial e-cigarette aerosols compared to control or vehicle. Kaur et al. conclude that immune cells are most affected, with the greatest dysregulation found in myeloid cells exposed to tobacco-flavored e-cigs and lymphoid cells exposed to fruit-flavored e-cigs. The up-and-down-regulated genes are heavily associated with innate immune response. The authors suggest that a Ly6G-deficient subset of neutrophils is found to be increased in abundance for the treatment groups, while gene expression remains consistent, which could indicate impaired function. Increased expression of CD4+ and CD8+ T cells along with their associated markers for proliferation and cytotoxicity is thought to be a result of activation following this decline in neutrophil-mediated immune response.

Strengths:

(1) Single-cell sequencing data can be very valuable in identifying potential health risks and clinical pathologies of lung conditions associated with e-cigarettes considering they are still relatively new.

(2) Not many studies have been performed on cell-type specific differential gene expression following exposure to e-cig aerosols.

(3) The assays performed address several factors of e-cig exposure such as metal concentration in the liquid and condensate, coil composition, cotinine/nicotine levels in

serum and the product itself, cell types affected, which genes are up- or down-regulated and what pathways they control.

(4) Considerations were made to ensure clinical relevance such as selecting mice whose ages corresponded with human adolescents so that the data collected was relevant.

We thank the reviewer for identifying the key strengths of our work and listing it in a concise and well-rounded fashion.

Weaknesses:

The exposure period of 1 hour a day for 5 days is not representative of chronic use and this time point may be too short to see a full response in all cell types. The experimental design is not well-supported based on the literature available for similar mouse models.

This study was not designed to study the effects of chronic exposures on lung tissues. We were interested in delineating the effect of acute exposures for which the proposed study design was chosen. Previous work by our group has performed similar exposures and has been well received by the community. We understand that chronic exposures will be interesting to look at, however that was not the purpose of this pilot study. We will now explicitly mention this aspect in the revised manuscript.

Several claims lack supporting evidence or use data that is not statistically significant. In particular, there were no statistical analyses to compare results across sex, so conclusions stating there is a sex bias for things like Ly6G+ neutrophil percentage by condition are observational.

We thank the reviewer for this observation, and we will include the necessary validations and details of the sex-based statistical analyses in the revised version of this manuscript.

Statistical analyses lack rigor and are not always displayed with the most appropriate graphical representation.

We thank the reviewer and will include all the necessary statistical details with more details in the revised manuscript.

Overall, the paper and its discussion are relatively limited and do not delve into the significance of the findings or how they fit into the bigger picture of the field.

We are in process of performing a few validity experiments and intend to include few other pieces of data to this manuscript to add to the overall merit of our findings. However as pointed out by the reviewer themselves the strength of this work is in the first ever scRNA seq analyses of mouse exposed to differently flavored e-cig aerosols in vivo. We also show cell-specific differential gene expression and address some of the major queries made around e-cig research including release of metals on a day-to-day basis from the same coil. The limited sample number make it difficult to draw solid conclusions from this work, which has been discussed as a shortcoming. However the major strength of this work is not in identifying specific trends but rather to explore the possible cell and gene targets to expand the study for longer (chronic) exposures with a larger sample group.

The manuscript lacks validation of findings in tissue by other methods such as staining.

We are conducting some studies and will include the validity experiments and staining in the revised manuscript to support our findings.

This paper provides a foundation for follow-up experiments that take a closer look at the effects of e-cig exposure on innate immunity. There is still room to elaborate on the differential gene expression within and between various cell types.

We thank the reviewer for this observation. The cell numbers for some cell clusters (especially epithelial cells) were too low. So, though we have performed the differential gene expression analyses on all the cell clusters, we refrained from discussing it in the manuscript to avoid over interpretation of our results. Only clusters with high enough (~1000) cells per sex per group were used to plot the heatmaps. We will also include the cell numbers for each cell type in the revisions to allow better interpretation of our data. Furthermore, the raw data from this study will be freely available to the public upon publication of this manuscript. This would enable the interested readers to access the raw data and study the cell types of interest in detail based on their study requirements. This data will be a useful resource for all in this community to inform and design future studies.

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